Biomedicine & Pharmacotherapy 188 (2025) 118182

Contents lists available at ScienceDirect

Biomedicine & Pharmacotherapy

journal homepage: www.elsevier.com/locate/biopha

Review

The bile acid-gut microbiota axis: A central hub for physiological regulation and a novel therapeutic target for metabolic diseases

Min Zhao, Jiafeng Zhao, Huimin Yang, Zirou Ouyang, Chang Lv, Zijun Geng, Jianhong Zhao *

Hebei Provincial Center for Clinical Laboratories, The Second Hospital of Hebei Medical University, Shijiazhuang, Hebei, China

ARTICLE INFO

Keywords:
Bile acid
Enterohepatic circulation
Gut microbiota
Metabolic disease
Treatment

ABSTRACT

Bile acids are a family of signaling molecules synthesized in the liver and metabolized by gut bacteria. As metabolites of the intestinal microbiota, bile acids bind to various receptors, and affect the metabolism and immune function of the host, including glucose and lipid metabolism, energy homeostasis, and inflammatory response. Conversely, bile acids also shape the composition of the gut microbiota. Given their critical role in physiological regulation, disrupted bile acid signaling is closely linked to metabolic diseases. Consequently, therapeutic strategies targeting bile acids are increasingly being explored. The size, composition, and function of the bile acid pool can be modulated through direct treatments (e.g., bile acid replacement therapy, administration of bile acid receptor agonists/antagonists) or indirect treatments (e.g., gut microbiota modulation, probiotic supplementation), providing new ideas for preventing and treating metabolic diseases.

1. Introduction

The gut microbiota and its metabolites, particularly bile acids (BAs), play crucial roles in various physiological processes, including nutrient absorption, metabolic regulation, physiological homeostasis, and host immune defense. BAs, a group of steroidal acids, can be classified into primary and secondary BAs based on their origin. Primary BAs, synthesized in the liver from cholesterol, include cholic acid (CA), chenodeoxycholic acid (CDCA), and their glycine or taurine conjugates. Upon entering the intestine, these primary BAs undergo microbial transformation into secondary BAs, such as deoxycholic acid (DCA) and lithocholic acid (LCA) [1]. These molecules modulate various host metabolic processes—including glucose and lipid metabolism, energy homeostasis, and inflammatory responses—by activating specific receptors in both hepatic and intestinal tissues.

Recent advances in elucidating the intricate interplay between the gut microbiota and BAs have revealed that compositional and functional alterations in the gut microbial community can remarkably influence the metabolism and composition of BAs, thereby contributing to the pathogenesis and progression of metabolic disorders [2,3]. Environmental perturbations that disrupt the gut microbiota or microbiota-host interactions can hinder the synthesis and metabolism of BAs. Dysregulation of BA signaling is associated with various metabolic disorders, including obesity, type 2 diabetes mellitus (T2DM), non-alcoholic fatty liver disease (NAFLD), inflammatory bowel disease (IBD), and cholestasis [4,5].

Given the physiological and pathological significance of BA signaling, BA-targeted therapies have emerged as promising treatment approaches for metabolic disorders. These strategies include direct interventions, such as BA replacement and administration of synthetic BA receptor agonists/antagonists, as well as indirect methods, such as fecal microbiota transplantation (FMT) and probiotic supplementation. The present review systematically explores three critical aspects of BA biology: (1) the signaling pathways associated with BA metabolism, (2) the bidirectional interactions between BAs and the gut microbiota and their implications in metabolic disorders, and (3) emerging BA-targeted therapeutic strategies. By integrating these interconnected elements, our work provides a comprehensive framework for understanding disease pathogenesis and informs the development of novel therapeutic interventions for metabolic diseases.

2. Synthesis and biotransformation of BAs

Hepatic conversion of cholesterol to BAs, mediated by cytochrome P450 enzymes, occurs through two distinct pathways: the classical pathway and the alternative pathway [6] (Fig. 1). Notably, the classical pathway contributes approximately 75 % of total BAs biosynthesis under physiological conditions. The liver is the exclusive site of the

* Corresponding author.
E-mail address: zhaojh_2002@hebmu.edu.cn (J. Zhao).

https://doi.org/10.1016/j.biopha.2025.118182
Received 16 April 2025; Received in revised form 18 May 2025; Accepted 21 May 2025
Available online 24 May 2025
0753-3322/© 2025 The Authors. Published by Elsevier Masson SAS. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

classical pathway for BA synthesis. Cholesterol is present in the both microsomes and cytoplasm of hepatocytes. The rate-limiting enzyme, cholesterol 7α-hydroxylase (CYP7A1), initiates the conversion of cholesterol to 7α-hydroxycholesterol. This intermediate is subsequently metabolized by 3β-hydroxyΔ5-C27-steroid dehydrogenase (HSD3B7) to form 7α-hydroxy-4-cholesten-3-one (C4) [7]. As a pivotal metabolic intermediate, C4 serves dual roles: (1) as the primary precursor for the synthesis of both CA and CDCA, and (2) as a clinically relevant biomarker for assessing the synthesis rates of hepatic BAs. C4 is further converted into the CA precursor (3α,7α,12α-trihydroxycholestanoic acid) under the action of sterol 12α-hydroxylase (CYP8B1), aldo-keto reductase (Δ4–3-oxosteroid-5β-reductase), and sterol 27-hydroxylase (CYP27A1) or into the CDCA precursor (3α,7α-dihydroxycholestanoic acid) in the absence of 12α-hydroxylation. Primary BAs are conjugated with taurine or glycine via two enzymatic steps: BA-coenzyme A (CoA) synthetase and BA-CoA:amino acid N-acyltransferase, forming conjugated primary BAs. These conjugated BAs then enter the intestine, where bile salt hydrolases (BSHs) deconjugated them, follow by 7α-dehydroxylation mediated by the bile acid-inducible (*bai*) operon of gut microbiota to generate DCA and LCA [8,9].

Following liver damage or impairment of the classical synthesis pathway in patients with liver diseases, the alternative pathway becomes activated as the primary route for BA synthesis [10]. The alternative pathway is initiated by two distinct enzymatic reactions: (1) sterol 25-hydroxylase hydroxylates cholesterol to form 25-hydroxycholesterol in the liver, and (2) CYP27A1 converts cholesterol into 3β-hydroxy-5-cholestenoic acid in hepatocytes, macrophages, and adrenal glands. Both compounds subsequently undergo oxysterol 7α-hydroxylase (CYP7B1)-mediated 7α-hydroxylation to yield 3β, 7α-dihydroxy-5-cholestenoic acid. This oxysterol is produced in extrahepatic tissues and transported to the liver, where it is converted into CDCA. Previous studies have demonstrated interspecies variation in CDCA conversion within liver microsomes among humans, mice, rats, dogs, and rabbits [8,9]. In mice, the majority of CDCA is rapidly converted into the more hydrophilic α-muricholic acid (α-MCA) by sterol-6β-hydroxylase. Subsequently, the 7α-hydroxy group of α-MCA undergoes 7α-β-isomerase-catalyzed epimerization to a 7β-hydroxy group, leading to the formation of β-MCA. In contrast, a portion of CDCA in humans is transformed into ursodeoxycholic acid (UDCA) through bacterial 7β-hydroxysteroid dehydrogenase (7β-HSDH) [11].

3. Enterohepatic circulation of BAs

Conjugated primary BAs, characterized by their strong hydrophilicity, are actively transported via the bile salt export pump (BSEP) and subsequently stored in the gallbladder [12]. Postprandial release of cholecystokinin from the pancreas stimulates the secretion of BAs into the intestinal lumen [13,14]. Approximately 95 % of conjugated BAs are actively absorbed by intestinal epithelial cells through the apical sodium-dependent bile acid transporter (ASBT). Following absorption, these conjugated BAs bind to ileal bile acid-binding protein (IBABP) and traverse the basolateral membrane via the organic solute transporter alpha/beta (OSTα/OSTβ) complex, ultimately returning to the liver via the portal vein. These recycled BAs, along with newly synthesized BAs from the liver, are secreted into the bile and subsequently released into the intestine during feeding, thereby completing the enterohepatic

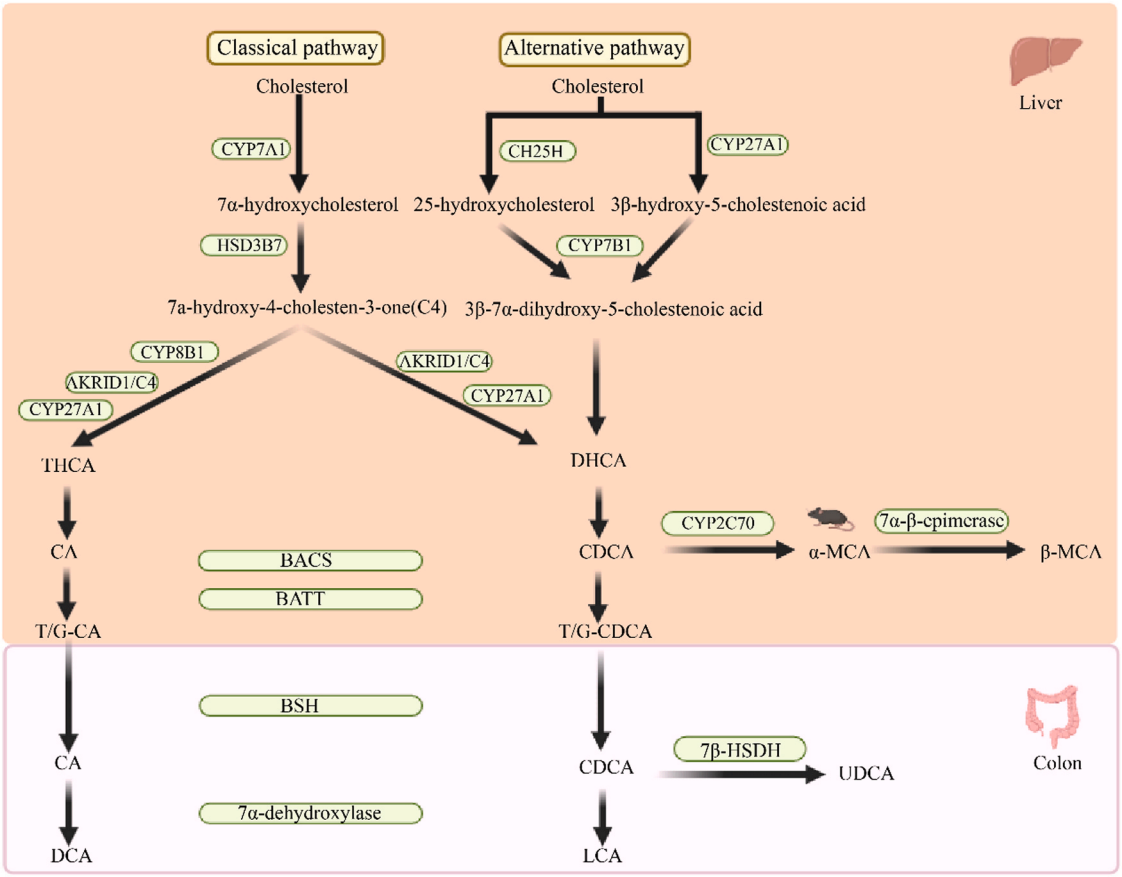


Fig. 1. Schematic diagram of bile acid synthesis and biotransformation. Note: The illustration depicts the synthesis pathways of primary bile acids (orange) in hepatocytes and secondary bile acids (pink) in the intestine. Primary bile acids CA and CDCA are initially synthesized from cholesterol in the liver via the classical pathway (mediated by CYP7A1) or the alternative pathway (mediated by CYP27A1), followed by extensive bacterial transformation in the intestine. These transformations ultimately result in a highly diverse bile acid pool.

circulation [9,15]. The remaining 5 % of unabsorbed BAs enter the colon, where gut microbiota mediate their biotransformation into secondary BAs via deconjugation and dehydroxylation processes [9]. The details are shown in Fig. 2.

4. Bile acid-gut microbiota interactions

4.1. Gut microbiota-mediated regulation of BA metabolism

During the enterohepatic circulation, microbial-synthesized enzymes play pivotal roles in BA transformation. This review focuses on BSHs, hydroxysteroid dehydrogenases (HSDHs), and the *bai* operon [9,16]. Table 1 summarizes the gut microbial species involved in BAs biosynthesis and their primary enzymatic functions.

4.1.1. BSH catalyzes the deconjugation reaction

Upon entering the intestine, conjugated BAs are hydrolyzed by microbiota-derived BSHs [17]. BSHs recognize and cleave the amide bond between BAs and glycine/taurine at their active sites, converting conjugated BAs into free forms (e.g., CA and CDCA) and corresponding amino acids (glycine or taurine), thus completing the deconjugation process [32,33]. Multiple bacterial genera, including *Bacteroides*, *Clostridium*, *Lactobacillus*, *Bifidobacterium*, *Enterococcus*, and *Listeria*, exhibit BSH activity [18,21,23,34]. These microbial species typically show enhanced bile tolerance and superior intestinal colonization capabilities [20]. The unconjugated BAs, characterized by their poor solubility, escape active absorption by small intestinal cells and transit to the colon, where they undergo further microbial metabolism to generate secondary BAs (e.g., DCA and LCA) [22]. Beyond deconjugation, BSHs also facilitates the formation of bacterial BA amides by catalyzing the conjugation of amines with BAs, thereby expanding the diversity of BAs [35,36].

4.1.2. HSDH catalyzes both hydroxyl oxidation and epimerization reactions

HSDH catalyzes the reversible oxidation of hydroxyl groups (-OH) to keto groups at specific positions (e.g., C-3, C-7, and C-12) of BAs [24,25]. Subsequently, HSDH alters the stereochemical configuration of BA molecules through redox reactions by inducing epimerization of hydroxyl groups at specific carbon atoms, thereby modifying the spatial structure of BAs [37]. Epimerization requires the coordinated action of two position-specific HSDHs (e.g., α- and β-HSDHs) [26]. For instance, 3α-HSDH, 7α-HSDH, or 12α-HSDH first oxidize hydroxyl groups to keto groups (e.g., 3-, 7-, 12-keto BAs), followed by reduction of these keto groups to β-hydroxyl groups by 3β-HSDH, 7β-HSDH, or 12β-HSDH, respectively [27]. Multiple gut bacteria species, including *Bacteroides*, *Clostridium*, and *Eubacterium* species, can independently or synergistically participate in this process [20,25,30]. Following epimerization, BA isomers with distinct spatial configurations are generated, including

Table 1
Bacterial bile acid transformations.

Reactions	Enzymes catalyzing bio-transformation reactions	Microbial producers
Deconjugation	BSH	*Turicibacter*[17]; *Bacteroides*[18]; *Lactobacillus*[19]; *Parabacteroides* [20]; *Bifidobacterium*[21]; *Clostridium*[21]; *Enterococcus*[22]; *Listeria*[23]
7-dehydroxylation	*bai* operon	*Clostridium*[20]; *Eubacterium*[20]; *Lachnospiraceae*[22]; *Ruminococcaceae*[22]; *Peptostreptococcaceae*[24]
Oxidation and epimerization	3α-HSDH	*Blautia*[25]
	3β-HSDH	*Parabacteroides merdae*[26]; *Odoribacteraceae*[26]; *Blautia*[25]
	7α-HSDH	*Escherichia*, *Fusobacterium*, *Blautia*, *Ruminococcus*, *Clostridium*[27]; *Parabacteroides*[20]; *Turicibacter* [17]
	7β-HSDH	*Blautia*[25]
	12α-HSDH	*Eggerthella*[28,29]; *Clostridium scindens*[30]; *Peptacetobacter hiranonis*[30]
	12β-HSDH	*Clostridium paraputrificum*[31]

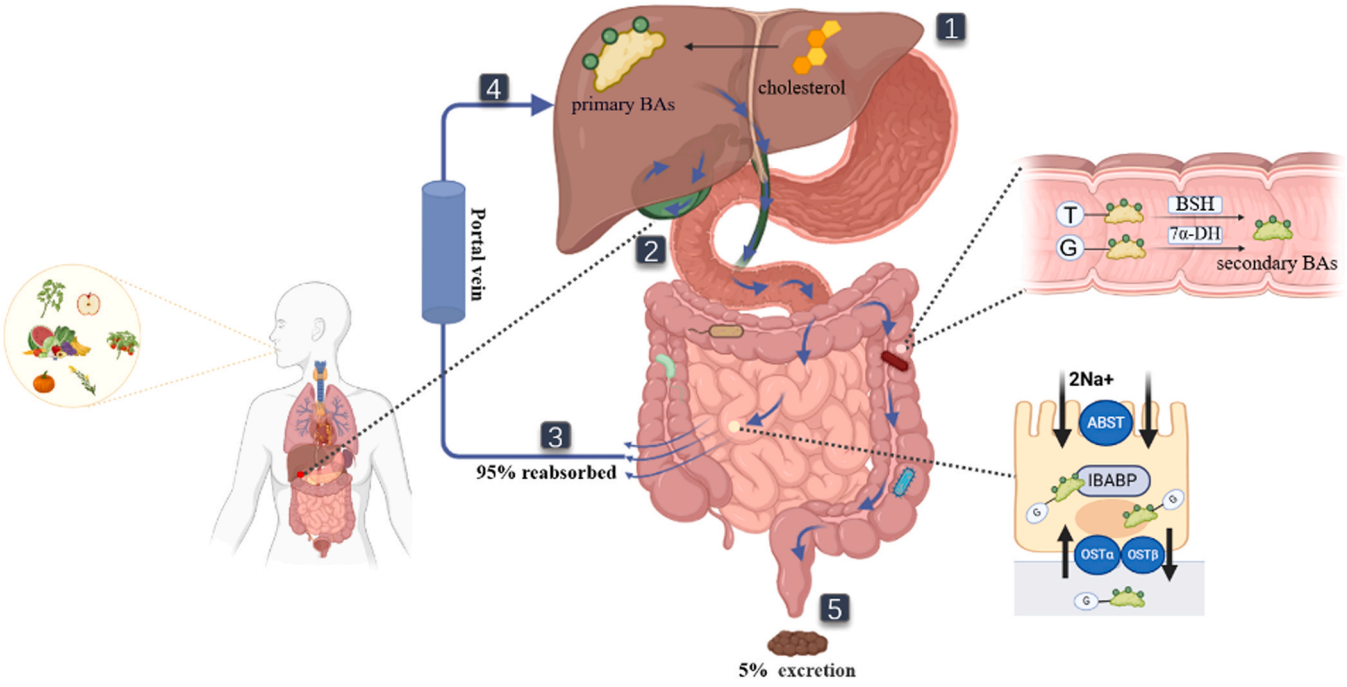


Fig. 2. Enterohepatic circulation of bile acids. Note: Numbers 1–5 and circular arrows indicate the flow pathway of bile acids among the liver, gallbladder, and intestine. G, glycine-conjugated; T, taurine-conjugated. Approximately 95 % of bile acids reaching the terminal ileum undergo hepatic recirculation, while the remaining 5 % that are not reabsorbed enter the colon and are transformed into secondary bile acids under the catalysis of gut microbiota.

isodeoxycholic acid (iso-DCA), UDCA, and epideoxycholic acid [25,27,31], which subsequently modulate host BA metabolism and gut microbiota function.

4.1.3. The bai operon catalyzes the dehydroxylation reaction

The *bai* operon is a gene cluster encoding multiple enzymes, including BaiB, BaiCD, BaiA2, BaiE, BaiF, BaiG, BaiH, and BaiI, which collectively catalyze BA dehydroxylation [24]. Initially, BaiB, functioning as a bile acid-CoA ligase, catalyzes the adenylation of BAs to form cholyl-coenzyme A (cholyl-CoA), representing a critical step in the overall reaction cascade [30]. Subsequently, BaiA2 catalyzes 3-dehydroxylation of cholyl-CoA to generate 3-oxo-cholyl-CoA, which is further oxidized by BaiCD to form 3-oxo-4,5-dehydro-cholyl-CoA [34]. Next, BaiE catalyzes the critical step of 7α-dehydroxylation to form 3-oxo-Δ4,5–6,7-deoxycholyl-CoA [37]. During the reduction phase, BaiH removes the CoA moiety. Two sequential oxidation steps mediated by BaiCD and BaiA2, ultimately yielding DCA or other secondary BAs [38]. *Firmicutes* genera, including *Clostridium* and *Eubacterium*, utilize the *bai* operon to mediate 7α-dehydroxylation, converting CA and CDCA into DCA and LCA, respectively [20,37].

4.2. BAs modulate gut microbiota composition

A bidirectional regulatory relationship exists between the gut microbiota and BAs. On the one hand, the gut microbiota modulates the metabolism of BAs, influencing the size and composition of the BA pool [9]. On the other hand, via their unique chemical properties and physiological functions, BAs selectively promote or inhibit the growth of specific microbial populations, thereby reshaping gut microbiota composition [22]. The following sections detail the regulatory effects of BAs on the gut microbiota.

4.2.1. Direct antimicrobial activity of BAs

BAs can act as antimicrobial agents by disrupting cell membranes or altering intracellular macromolecular structures [39]. At low concentrations, BAs modulate membrane permeability and hydrophobicity through interactions with microbial surface proteins, whereas at high concentrations, they rapidly degrade membrane proteins in microbial cells [40]. Therefore, only those microbial communities that can tolerate high concentrations of BAs survive in the intestinal environment. Previous research indicates that different types of BAs exhibit remarkable inhibitory effects on specific pathogens. For instance, DCA inhibits the growth of *Clostridioides difficile*, while UDCA directly suppresses the germination of *C. difficile* spores *in vitro* [41]. Additionally, iso-LCA demonstrates substantial antimicrobial activity against Gram-positive multidrug-resistant pathogens, such as *Enterococcus faecium* [26]. Sannasiddappa et al. [42] revealed that BAs exert antibacterial effects on *Staphylococcus aureus* at different concentrations through distinct mechanisms. At the minimum inhibitory concentration, BAs achieve bactericidal effects by disrupting the bacterial cell membrane, leading to the leakage of cellular contents [40]. In contrast, at subinhibitory concentrations, BAs inhibit bacterial growth by impairing the proton motive force in bacterial cells, thereby disrupting the transmembrane potential and increasing membrane permeability. Notably, unconjugated BAs passively penetrate bacterial cell membranes via the “flip-flop” mechanism, with their efficiency affected by the number of hydroxyl groups in the BA molecules: the fewer hydroxyl groups, the greater the penetration ability [40]. Moreover, unconjugated BAs exhibit stronger antimicrobial activity compared to their conjugated counterparts [42,43].

4.2.2. BA-mediated regulation of gut microbiota structure

Studies have shown that supplemental BAs can alter gut microbiota composition [9]. Obeticholic acid (OCA), a derivative of CDCA, inhibits endogenous BA synthesis and increases the abundance of Gram-positive bacteria such as *Streptococcus thermophilus*, *Lactobacillus casei*, *Lactobacillus paracasei*, *Bifidobacterium breve*, and *Lactococcus lactis*, while reducing the abundance of Gram-negative bacteria [44]. Chen et al. [45] utilized the db/db mice model for assessing the effects of BAs on obesity and T2DM; the authors found that the administration of glycoursodeoxycholic acid (GUDCA) effectively mitigated the onset and progression of diabetes. Additionally, the abundance of *Bacteroides vulgatus* in the intestines of db/db mice was significantly increased [46]. Similarly, supplementation with DCA also induces significant alterations in gut microbiota composition. DCA administration promotes the growth of *Akkermansia muciniphila* in the intestines of mice [46].

4.2.3. FXR activation-dependent indirect modulation of the gut microbiota by BAs

In addition to direct regulation, BAs can indirectly modulate gut microbiota composition by activating specific receptors such as the farnesoid X receptor (FXR) [15]. FXR activation upregulates the expression of genes related to small intestinal mucosal defense, such as nitric oxide synthase and interleukin-18, thereby promoting the synthesis of antimicrobial molecules such as nitric oxide and inhibiting bacterial growth [47].

5. Physiological regulation of BAs and their receptor functions

BAs function as critical signaling molecules by binding to various receptors, including the FXR [48], Takeda G-protein-coupled receptor 5 (TGR5) [45], vitamin D receptor [49], and pregnane X receptor [50], thereby modulating downstream cellular signaling pathways.

The investigation of the expression and functional changes of BAs and their receptors in metabolic diseases provides critical insights into the pathogenesis of these conditions. This review focuses on the roles of FXR and TGR5 in the synthesis and metabolism of BAs (Fig. 3) and futher discusses how these regulatory processes influence systemic glucose and lipid metabolism (Fig. 4), energy homeostasis, and inflammatory responses (Fig. 5).

5.1. Regulation of FXR and TGR5 in BA homeostasis and metabolic synthesis

5.1.1. FXR

FXR is expressed in multiple organs, including the liver, intestine, kidneys, white adipose tissue, and immune cells [51], with the highest expression levels observed in the liver and intestine. As the primary receptor for BAs, FXR plays a pivotal role in regulating the synthesis, transport, and metabolism of BAs [52]. Primary BAs (CA and CDCA) and secondary BAs (DCA and LCA) act as activators of FXR, and their activation efficacy follows the order: CDCA > DCA > CA > LCA [53]. In contrast, T-αMCA and T-βMCA function as intestinal FXR antagonists [54]. Following activation by BAs, hepatic FXR induces the expression of the small heterodimer partner (SHP). SHP synergizes with hepatocyte nuclear factor-4α (HNF-4α) and liver receptor homolog-1 (LRH-1) to suppress the expression of key enzymes in BA biosynthesis, including *cyp7a1* and *cyp8b1* [55]. This process promotes the efficient excretion of BAs. In the intestine, BAs activate FXR, inducing the production of fibroblast growth factor 15/19 (FGF15/19). This signaling molecule is then transported to the liver, where it specifically binds to fibroblast growth factor receptor 4 (FGFR4) and β-klotho. This interaction activates the downstream ERK/JNK signaling pathway, which subsequently downregulates the expression of hepatic *cyp7a1* and reduces BA synthesis. This pathway forms a critical gut-liver axis, establishing feedback inhibition of BA synthesis [56].

Additionally, FXR modulates the function of various transport proteins, thereby influencing the synthesis of BAs. In the liver, the BSEP mediates the excretion of BAs. ATP-binding cassette G5/G8 facilitates cholesterol efflux, and multidrug resistant protein 2/3 regulates phospholipid secretion, collectively ensuring the proper assembly and excretion of bile components. FXR maintains low levels of BAs in hepatocytes by suppressing the activity of sodium taurocholate

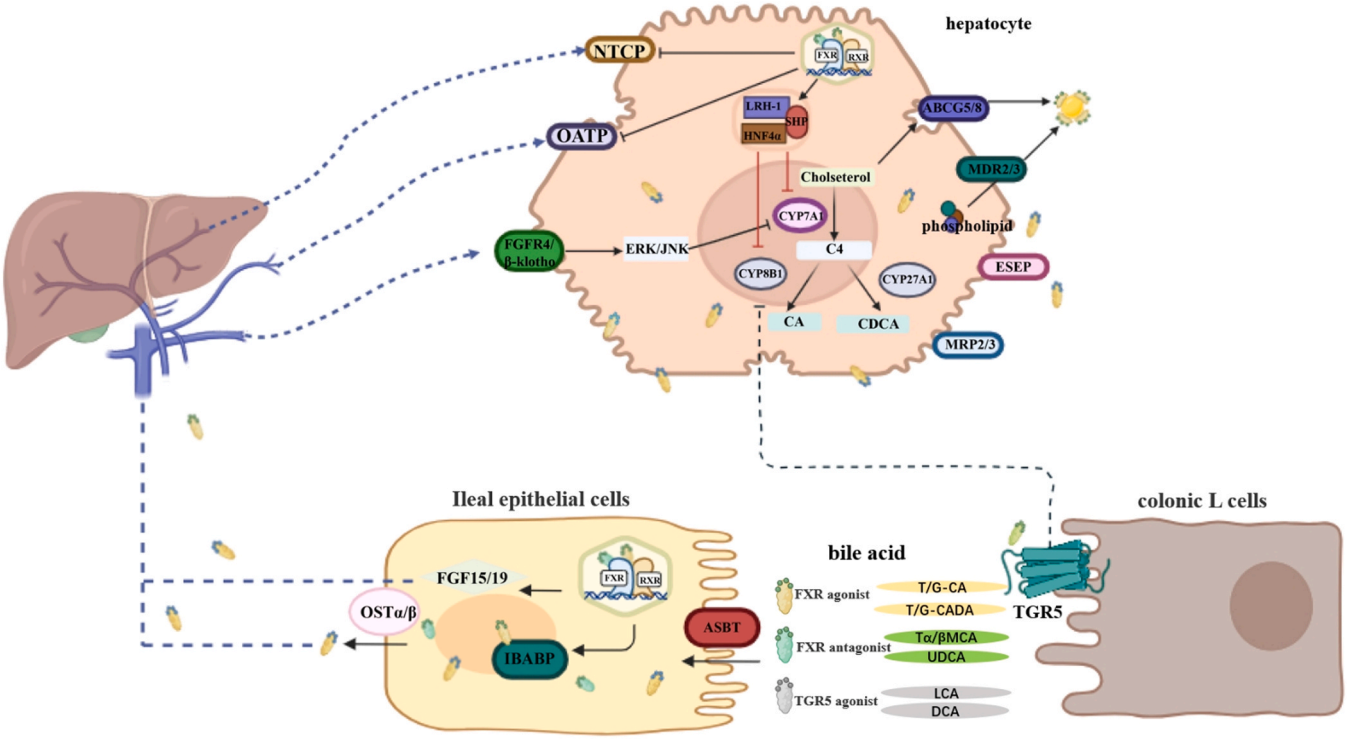


Fig. 3. Regulation of bile acid receptors in bile acid homeostasis and synthesis metabolism. Note: The nuclear receptor FXR and membrane receptor TGR5 regulate bile acid synthesis and metabolism. Bile acids are transported back to the liver via the enterohepatic circulation through several active transporters on ileal cells (ASBT, OSTα/β) and hepatocytes (NTCP, OATP, and the FGFR4/β-klotho complex). T/G-CA and T/G-CDCA are agonists of FXR (highlighted in yellow), LCA and DCA are activators of TGR5 (highlighted in gray), and T-α/βMCA and UDCA are antagonists of FXR (highlighted in green). Bacterially transformed bile acids activate different bile acid receptors, modulating signaling pathways and extensively covering a complex symbiotic metabolic network.

cotransporting polypeptide (NTCP) and organic anion transport polypeptide (OATP) on the sinusoidal membrane. In the intestine, FXR promotes the transport of BAs to the sinusoidal membrane by inhibiting ASBT expression and enhancing IBABP expression. Under the action of the OSTα/β complex, BAs are secreted into the bloodstream, which prevents their accumulation in enterocytes and maintains intestinal barrier function [56,57].

5.1.2. TGR5

TGR5 is a membrane-bound receptor widely distributed in the liver, intestine, pancreas, central nervous system, and immune cells [58]. BAs show varying efficacy in activating TGR5, with the potency in the following order: LCA > DCA > CDCA > CA [59]. TGR5 activation downregulates *cyp8b1* expression, leading to a decrease in the ratio of 12α-hydroxylated BAs (12α-OH BAs) to non-12α-hydroxylated BAs (non-12α-OH BAs) in the BA pool [60]. Additionally, TGR5 enhances the efficiency of the synthesis and metabolism of BAs by activating other enzymes in the alternative pathway, such as CYP7B1 and CYP27A1 [61].

5.2. Role of BAs in regulating blood glucose levels

As crucial bioactive molecules, BAs actively participate in the regulation of blood glucose homeostasis by effectively inhibiting hepatic gluconeogenesis, promoting glycogen synthesis, and enhancing insulin sensitivity, thereby maintaining glucose stability. In the liver, the activated FXR induces SHP expression. SHP subsequently suppresses key gluconeogenic enzymes, such as phosphoenolpyruvate carboxykinase (PEPCK), fructose-1,6-bisphosphatase (FBPase), and glucose-6-phosphatase (G6Pase), through SHP-dependent interactions with HNF-4 [62] and forkhead transcription factor (FOXO1) [63], thereby inhibiting hepatic gluconeogenesis [64–66]. FXR also promotes hepatic glycogen synthesis by activating the PI3K/AKT/GSK3β signaling cascade, which upregulates the expression of glycogen synthase [67]. Additionally, FXR enhances the function of glucose transporter 2 (GLUT2), which facilitates the transport of blood glucose into hepatocytes, stimulating insulin production and secretion [68]. Intestinal FXR induces the production of FGF15/19, which, following transportation to the liver, binds to the FGFR4/β-klotho complex. This process, independent of insulin signaling, directly promotes hepatic glycogen synthesis and suppresses gluconeogenesis [69]. Moreover, intestinal BAs activate TGR5, stimulating the secretion of glucagon-like peptide-1 (GLP-1) [70]. GLP-1 regulates glucose metabolism and energy balance by promoting insulin production, inhibiting gastric emptying and acid secretion, delaying intestinal motility, enhancing satiety, and reducing food intake [71,72].

5.3. Role of BAs in regulating lipid metabolism

BAs play a crucial role in regulating lipid metabolism and maintaining lipid homeostasis. On the one hand, BAs promote fat emulsification by binding to lipids and disintegrating them into smaller particles [73]. BAs increase the contact area between fats and lipases, which accelerates lipid digestion and absorption. BAs also bind to cholesterol transport proteins in the intestine, which facilitates cholesterol excretion and reduces its reabsorption [74]. On the other hand, BAs activate receptors, particularly FXR, and regulate downstream pathways, thereby achieving precise control over lipid metabolism [75]. Following the binding of BAs to FXR, the activated FXR induces SHP expression, which suppresses the transcription of the sterol regulatory element-binding protein 1c (SREBP-1c). This further downregulates the activity of key lipogenic enzymes, such as fatty acid synthetase (FAS), acetyl-CoA carboxylase (ACC), and stearoyl-CoA desaturase-1 (SCD-1), thereby reducing hepatic triglyceride levels [76,77]. Simultaneously, FXR reduces lipogenesis by enhancing the expression of peroxisome

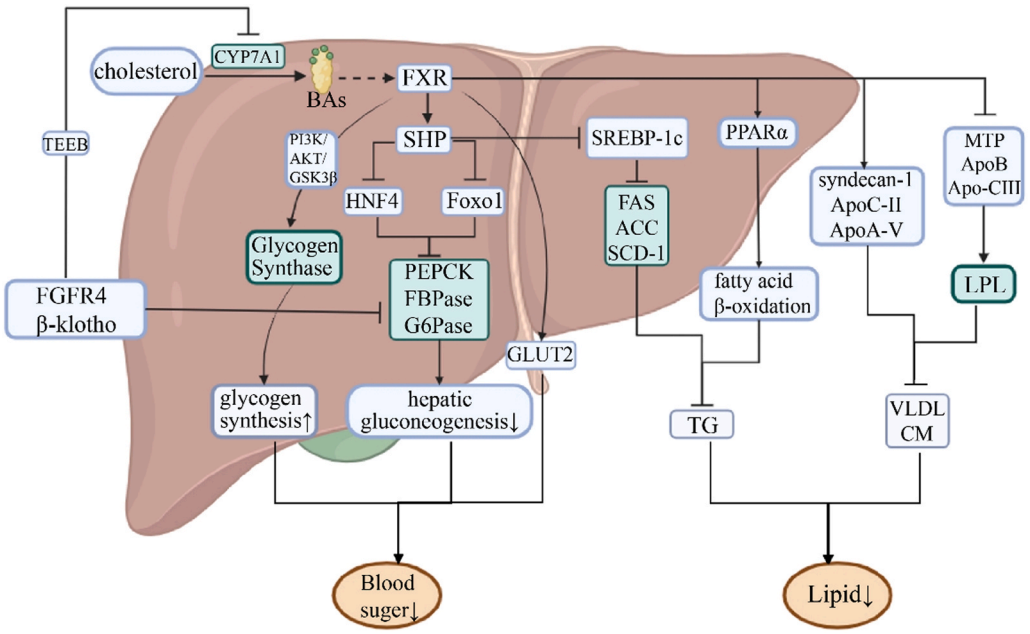


Fig. 4. Regulatory effects of bile acids and their receptors on systemic glucose and lipid metabolism. Note: In the liver, activated FXR induces the expression of SHP, which in turn suppresses the expression of key gluconeogenic enzymes including PEPCK, FBPase, and G6Pase through an SHP-dependent mechanism involving the inhibition of HNF4 and FOXO1. Concurrently, FXR activates the PI3K/AKT/GSK3β signaling pathway to promote glycogen synthesis and enhances the function of glucose transporter GLUT2, thereby improving hepatic glucose uptake. Intestinal FXR stimulates the production of FGF15/19, which upon reaching the liver binds to the FGFR4/β-klotho complex to directly promote glycogen synthesis while suppressing gluconeogenesis. Regarding lipid metabolism, FXR-mediated SHP induction inhibits the expression of SREBP-1c and its downstream lipogenic enzymes including FAS, ACC, and SCD-1. Simultaneously, FXR upregulates PPARα to enhance fatty acid β-oxidation. Furthermore, FXR modulates lipid metabolism by activating syndecan-1, ApoC-II, and ApoA-V, while suppressing MTP, ApoB, and ApoC-III expression, collectively reducing hepatic VLDL and chylomicron levels.
SHP, Small heterodimer partner; PEPCK, phosphoenolpyruvate carboxykinase; FBPase, fructose-1,6-bisphosphatase; G6Pase, glucose-6-phosphatase; HNF4, Hepatocyte nuclear factor 4; FOXO1, forkhead transcription factor; GLUT2, Glucose transporter; SREBP-1c, sterol regulatory element-binding protein 1c; FAS, fatty acid synthetase; ACC, acetyl-CoA carboxylase; SCD-1, stearoyl-CoA desaturase-1; PPARα, peroxisome proliferator-activated receptor α; Apo, apolipoprotein; MTP, microsomal triglyceride transfer protein; VLDL, very-low-density lipoprotein.

proliferator-activated receptor α (PPARα), which promotes fatty acid β-oxidation [78]. FXR activation upregulates syndecan-1[79], apolipoprotein C-II (ApoC-II) [80], and apolipoprotein A-V (ApoA-V) [81]. Conversely, it suppresses microsomal triglyceride transfer protein (MTP) [82], apolipoprotein B (ApoB) [82], and apolipoprotein C-III (ApoC-III) [83], reducing hepatic VLDL and chylomicron levels. This mechanism contributes to the stabilization of plasma cholesterol levels and facilitates hyperlipidemia prevention.

5.4. Role of BAs in regulating energy metabolism

BAs modulate the expression of energy metabolism-related genes through receptor binding. In the intestine, TGR5 activation stimulates the cAMP-dependent thyroid hormone-activating enzyme deiodinase-2 (DIO2), which converts inactive thyroxine (T4) into active triiodothyronine (T3) [84]. Activated T3 enhances mitochondrial oxygen consumption and energy expenditure in murine brown adipocytes and human skeletal muscle cells. In muscle and brown adipose tissue, activated TGR5 prevents obesity and its complications by enhancing insulin sensitivity and promoting energy expenditure. Additionally, intestinal FXR enhances fatty acid β-oxidation and white adipose tissue browning by inducing the expression of proteins such as peroxisome proliferator-activated receptor gamma coactivator 1α/β (PGC1α and PGC1β) [85] and uncoupling protein 1 (UCP1) [86].

5.5. Role of BAs in modulating inflammatory responses

Following their entry into the intestine, liver-secreted BAs are converted by gut microbiota from primary BAs to secondary BAs, which participate in the regulation of immune cell differentiation and subsequently modulate inflammatory responses [87]. Certain secondary BAs, such as 3-oxo-LCA and iso-LCA, regulate immune cell balance by promoting Treg cell differentiation and inhibiting Th17 cell differentiation, thereby enhancing the anti-inflammatory capacity of the immune system [88,89]. This balance is crucial for maintaining organismal health, resisting pathogen infections, and preventing the onset of autoimmune diseases. Additionally, BAs-activated cAMP signaling inhibits the NF-κB pathway, thereby suppressing the expression of lipopolysaccharide (LPS)-induced inflammatory cytokines [90]. FXR and TGR5 are expressed in various immune cells, this expression may contribute to their anti-inflammatory effects [91]. A previous study demonstrates that TGR5 suppresses LPS-induced chemokine expression via the AKT-mTOR-LIP pathway, thereby ameliorating obesity-induced insulin resistance and inflammatory responses [92].

6. Regulatory mechanisms mediated by the BA-gut microbiota axis in metabolic diseases

The pathogenesis of abnormal metabolic conditions, such as obesity, T2DM, NAFLD, IBD, and cholestasis, is frequently associated with metabolic dysregulation. These conditions are characterized by chronic low-grade inflammation, marked by the recruitment of immune-inflammatory cells and abnormal production of cytokines and acute-phase reactants. Some researchers have defined these disorders as “meta-inflammatory disorders” [93,94]. Given the crucial role of BAs in regulating glucose and lipid metabolism and maintaining systemic energy homeostasis, it is increasingly recognized that abnormal alterations in BA levels and composition may serve as crucial driving factors in the pathogenesis of these metabolic disorders. The following discussion will focus on the roles of BAs and their microbial transformations in

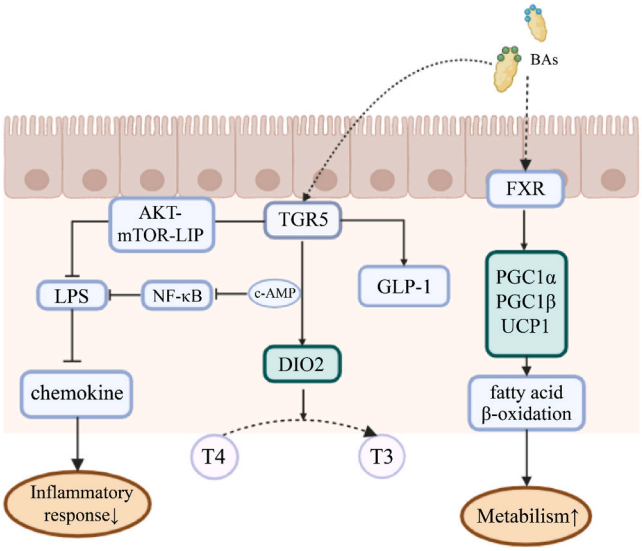


Fig. 5. Regulatory effects of bile acids and their receptors on energy metabolism and inflammatory responses. Note: In the intestine, bile acids activate the TGR5 receptor, which triggers the cAMP signaling pathway to promote the conversion of T4 into T3 by DIO2. Activated T3 significantly enhances mitochondrial oxygen consumption in murine brown adipocytes and human skeletal muscle cells, thereby increasing energy expenditure. Concurrently, activation of the intestinal FXR upregulates the expression of PGC1α/PGC1β and UCP1, enhancing fatty acid β-oxidation and promoting white adipose tissue browning, ultimately improving systemic energy metabolism. Bile acids inhibit NF-κB signaling via cAMP, reducing inflammation caused by LPS. The TGR5 receptor also inhibits LPS-induced chemokine production via the AKT-mTOR-LIP signaling pathway. Together, these mechanisms constitute the molecular basis of bile acid-mediated regulation of energy metabolism and inflammatory responses.
T4, inactive thyroxine; T3, active triiodothyronine; DIO2, deiodinase-2; PGC1α/PGC1β, peroxisome proliferator-activated receptor gamma coactivator 1α and 1β; UCP1, uncoupling protein 1; LPS, lipopolysaccharide.

obesity-related disorders, NAFLD, IBD, and cholestasis, with an aim to provide new insights for the prevention and treatment of these metabolic diseases.

6.1. Obesity and T2DM

T2DM is one of the leading causes of mortality and disability worldwide. According to previous studies, 49.6 % of the increase in T2DM prevalence is attributable to obesity trends [95]. In obese individuals, the abundance or increased number of adipocytes reduce insulin receptor density on cell membranes, leading to insulin resistance. As insulin resistance progresses, it ultimately leads to the development of diabetes.

Patients with obesity and insulin resistance show increased synthesis of BAs, particularly elevated serum levels of 12α-OH BAs (including CA and its derivatives) and significantly reduced levels of non-12α-OH BAs (including CDCA and its derivatives) [96]. A previous study demonstrated that mice lacking the CYP8B1 enzyme (*cyp8b1*$^{-/-}$) cannot synthesize CA and exhibit increased production of CDCA. These mice were resistant to high-fat diet (HFD)-induced obesity and showed improved insulin sensitivity [97]. *cyp8b1*$^{-/-}$ mice also exhibited elevated levels of T-MCA, an intestinal FXR antagonist, which reduced FGF15 production and thereby influenced the synthesis and metabolism of BAs. Conversely, overexpression of the *cyp8b1* gene in mice exacerbates metabolic issues such as SREBP-1c-mediated lipogenesis, dyslipidemia, and insulin resistance [98]. The aforementioned studies demonstrate that CYP8B1 regulates the proportion of 12α-OH BAs within the BA pool, determining whether the metabolic end products are CA or CDCA.

Alterations in the proportion of 12α-OH BAs can significantly reshape gut microbiota composition and play a crucial role in systemic metabolic regulation [99]. In an HFD-induced mouse model, researchers observed a reduction in the abundance of *Clostridium scindens* in the gut microbiota, which positively correlated with UDCA and LCA levels. Supplementation with *C. scindens* or UDCA increased the levels of hepatic non-12α-OH BAs, thereby alleviating HFD-induced obesity [100]. Gou et al. [101] used machine learning to identify variations in gut microbiota composition in T2DM patients and discovered 14 microbial species predictive of T2DM risk. The abundance of *Akkermansia*, *Bacteroides*, *Bifidobacterium*, *Faecalibacterium*, and *Roseburia* was reduced in T2DM patients, while the abundance of *Blautia*, *Fusobacterium*, and *Ruminococcus* showed a positive correlation with T2DM [102]. Interestingly, these bacteria exhibit the ability to metabolize BAs. Specifically, *Blautia* and *Ruminococcus* possess 3α-HSDH activity; *Bacteroides fragilis* demonstrates 7α-HSDH activity, *Ruminococcus gnavus* is a major producer of iso-DCA [103], and *Bifidobacterium* carries multiple BSH genes.

6.2. NAFLD

NAFLD includes a spectrum of diseases characterized by inflammation and hepatocyte injury, ranging from simple hepatic steatosis to progressive nonalcoholic steatohepatitis (NASH), liver fibrosis, and potentially advancing to cirrhosis and hepatocellular carcinoma [104]. NAFLD often coexists with other metabolic disorders such as obesity, T2DM, and cardiovascular diseases, thereby increasing the risk of adverse clinical outcomes [105].

Previous studies have confirmed dysregulation of BA homeostasis in NAFLD patients. Compared to the healthy control group, NASH patients showed higher levels of total BA, CA, and CDCA in their feces and an increased ratio of primary BAs to secondary BAs [106,107]. A study involving 178 NAFLD patients revealed a significant reduction in the serum levels of hyodeoxycholic acid (HDCA), which was negatively correlated with the presence and severity of NAFLD. Subsequent investigation using an NAFLD mice model demonstrated the therapeutic efficacy of HDCA [108]. The results showed that HDCA alleviated disease severity in NAFLD mice by upregulating hepatic *cyp7b1* expression. Additionally, HDCA supplementation significantly increased the abundance of beneficial gut bacteria such as *Parabacteroides distasonis* [108]. During NASH, the expression of BA-related receptor genes, such as *fxr*, is downregulated [109]. Clifford et al. [110] demonstrated that the activated FXR reduces hepatic triglyceride levels through two distinct mechanisms: BA-dependent reduction of intestinal lipid absorption and regulation of lipogenic gene expression. Moreover, FXR agonists, such as OCA, have demonstrated therapeutic effects in NAFLD mice and are currently under clinical investigation as potential treatments for NAFLD [111]. In a steatohepatitis mice model, supplementation with BAR502, a dual TGR5-FXR agonist, promoted white adipose tissue browning and reduced hepatic steatosis and fibrosis [112].

6.3. IBD

IBD, which includes Crohn's disease and ulcerative colitis, represents a group of clinically significant chronic inflammatory disorders of the gastrointestinal tract. The incidence and prevalence of IBD are increasing globally, and IBD has been reported to affect approximately 7 million people worldwide. Although the precise etiology of IBD remains unclear, previous studies have revealed alterations in the composition and function of the gut microbiota and changes in microbial metabolites, such as the BA pool, in IBD patients. Duboc et al. [113] demonstrated that IBD patients exhibit gut microbiota dysbiosis, with impaired microbial enzymatic functions such as deconjugation, transformation, and desulfation, leading to alterations in the BA pool composition. Alterations in BAs affect the expression of intestinal tight junction proteins and stem cell renewal, compromising the intestinal barrier and

exacerbating the incidence of IBD [114]. Recent studies have shown that reduced tolerance of intestinal epithelial cells to BAs is closely associated with elevated levels of tumor necrosis factor (TNF) during colitis development, with BAs and TNF synergistically inducing intestinal epithelial cell death [115]. Therefore, the pathogenesis of IBD may be associated with disruptions in the circulation and metabolism of BAs, suggesting that BAs have potential as a novel therapeutic strategy for IBD [116].

Studies have revealed a reduction in the diversity and abundance of the gut microbiota in IBD patients, particularly bacteria encoding BSH and the *bai* operon, including *Bacteroides*, *Clostridium*, *Lactobacillus*, *Bifidobacterium*, and *Listeria* [117,118]. Wilson et al. [119] demonstrated that IBD patients exhibit reduced activity of FXR and PXR. By synthesizing BA receptor agonists, direct stimulation of FXR exerted anti-inflammatory effects and provided protection in dextran sulfate sodium-induced colitis mice. Symptoms such as colon shortening, ulcer formation, and inflammatory cell infiltration were remarkably alleviated. However, supplementation with FXR agonists showed no effect in *fxr*$^{-/-}$ mice and instead resulted in exacerbation of intestinal inflammation and worsening of mucosal damage [120]. Additionally, the activated FXR stimulates the production of FGF15/19, which suppresses hepatic *cyp7a1* expression, regulates reabsorption of BAs, and prevents excessive accumulation of BAs within enterocytes, thereby maintaining BA homeostasis in the intestine [120]. In summary, FXR has potential therapeutic significance for diseases associated with intestinal barrier dysfunction and offers a promising avenue for future research and treatment strategies.

6.4. Cholestasis

Cholestasis is characterized by impaired bile flow, leading to excessive accumulation of BAs in the liver, reduced levels of intestinal BAs, and increased levels of systemic BAs. The accumulated BAs cause hepatocyte and biliary damage as well as inflammation. Over time, chronic cholestasis can progress to liver fibrosis, cirrhosis, and ultimately result in liver failure or hepatocellular carcinoma [121]. In patients with cholestasis, the expression of hepatobiliary transport proteins is upregulated, which reduces the uptake and synthesis of BAs and promotes their excretion, thereby limiting the accumulation of potentially toxic substances [122]. The adaptive regulatory mechanisms exhibited by the body under pathological conditions are mediated by multiple nuclear receptors, such as FXR and PXR, and intestinal hormones such as FGF-19. Activated FXR suppresses *cyp7a1* expression to reduce BA synthesis and inhibits NTCP and OATP activity to decrease hepatic BA uptake, thereby preventing BA accumulation within hepatocytes [122].

A recent study by Song et al. [123] revealed that hepatic *fgf4* is a direct target gene of FXR, which downregulates the transcription of *cyp7a1* and *cyp8b1* through paracrine signaling pathways. FXR-FGF4 controls BA synthesis and maintains BA pool homeostasis by activating the FGFR4-LRH-1 pathway. In cholestasis patients, hepatic FXR and FGF4 expression levels are reduced, which impairs the FXR-FGF4 signaling axis. Compared to wild-type mice, *fgf4*-deficient mice showed increased levels of BAs in the liver and intestine, resulting in the exacerbation of cholestasis and liver injury symptoms. Conversely, the administration of an FGF4 analog to mice suppresses BA synthesis and alleviates histopathological damage, such as necrosis and biliary fibrosis. In addition to affecting homeostasis of the BA pool, cholestasis also causes long-term effects on the intestinal flora. A previous study showed that BA stasis and liver injury typically recovered within a short period in a mouse model of hepatic cholestasis induced by temporary complete bile duct obstruction. However, the dysbiotic state of the gut microbiota did not recover accordingly, with long-term effects such as small intestinal bacterial overgrowth, reduced gut microbiota diversity, and elevated inflammation levels [124]. Studies have demonstrated that the pathogenesis of cholestasis is closely associated with gut microbiota dysbiosis, with cholestasis patients exhibiting significantly reduced fecal microbial diversity and decreased abundance of *Prevotella*, *Faecalibacterium*, *Bacteroides*, *Sutterella*, and *Oscillospira* [125,126]. In a mouse model of primary sclerosing cholangitis, *Prevotella* intervention significantly alleviated cholestasis symptoms by modulating the gut microbiota to influence the activity of receptors such as FXR, resulting in the suppression of BA synthesis and mitigation of hepatic cholestasis [126].

7. The BA-gut microbiota axis: a novel therapeutic target for metabolic diseases

Advances in genomics and metabolomics have revealed that the pathophysiology of metabolic diseases is regulated by BAs and their receptors. This discovery provides strong evidence for BA-based therapeutic strategies, and offers new perspectives and molecular targets for the diagnosis and treatment of metabolic diseases. Current BA-based therapies include direct agonists/antagonists targeting BA receptors and indirect modulation of the gut microbiota through the gut microbiota-BA interaction. Table 2 summarizes the roles of current BA-based therapies in metabolic diseases.

Note：LGG: Lactobacillus rhamnosus GG; IRa: insulin resistance; IRb: insulin receptor; BG: blood glucose; IRS-1: insulin receptor substrate 1; ALT: Alanine Aminotransferase; T-LCA: taurolithocholic acid; UCP-1: Uncoupling Protein 1. ↓: The down arrow indicates a decrease in expression level, content, etc. ↑: The up arrow indicates an increase in expression level, content, etc.

7.1. Gut microbiota-BA interaction: improving BA metabolism by modulating gut microbiota

7.1.1. FMT

With an improved understanding of the role of the gut microbiota in diseases, restoring gut microbiota homeostasis to alleviate disease symptoms has emerged as a promising research field, with the treatment of recurrent *C. difficile* infection as a widely applied area [143]. FMT, an effective approach for alleviating gut microbiota dysbiosis, has been utilized for treating and preventing various metabolic diseases. Allegretti et al. [127] observed that obese patients receiving FMT capsules from lean donors exhibited sustained changes in their gut microbiota and BA profiles, with fecal microbiota composition resembling that of lean donors, despite non-significant changes in body mass index. This finding demonstrates the efficacy of FMT in improving gut microbiota structure. Another clinical study showed that FMT enhanced BA metabolism by gut bacteria and delayed the progression of glucose tolerance impairment. Obese patients with FMT showed substantial improvements in blood glucose levels and insulin resistance, suggesting that FMT may help prevent metabolic syndrome [128]. These findings support FMT as a potential strategy for treating obesity and metabolic abnormalities.

7.1.2. Probiotic intervention

Zhang et al. [129] screened *Lactobacillus casei* YRL577 by measuring the BSH activity of 34 strains and confirmed its therapeutic effects on NAFLD. *L. casei* YRL577 significantly reduced the liver weight of NAFLD mice and regulated lipid metabolism, oxidative stress, and proinflammatory cytokine levels. Additionally, *L. casei* YRL577 modified genes in the intestinal bile acid pathway, including the upregulation of FXR and FGF15 and downregulation of ASBT mRNA levels, thereby contributing to the alleviation of NAFLD symptoms.

Modulation of the gut microbiota and inhibition of BA synthesis are potential strategies for treating cholestatic liver diseases, and multiple studies have confirmed the efficacy of probiotics in this context. Treatment with *Lactobacillus rhamnosus* GG (LGG) significantly alleviated liver inflammation, injury, fibrosis, and hepatic BA levels in mice with bile duct ligation (BDL). In BDL mice, the FXR antagonist T-βMCA was significantly increased, while the FXR agonist CDCA was reduced. LGG treatment reversed this trend, lowered T-βMCA level, and restored CDCA to normal levels. LGG treatment also significantly increased

Table 2
Mechanisms and related drugs/microorganisms for the treatment of metabolic diseases.

Mechanism	Related Drugs/ Microorganisms	Disease Treated	Therapeutic outcome	Subjects
Gut Microbiota	FMT	Obesity; Diabetes	IR[a], BG↓	Human[127,128]
	Lactobacillus casei YRL577	NAFLD	FXR、FGF15↑ ABST↓	Mice[129]
	LGG	Cholestasis	Bile acid excretion，CDCA↑ T-βMCA、CYP7A1、BAs synthesis↓	Mice[130]
	L. acidophilus	Cholestasis	CYP7A1、BAs synthesis↓ Bile acid excretion↑	Mice[131]
Receptor Agonists	OCA	NASH	bile acid synthesis↓ *Firmicutes*↑, inflammation response↓	Humans[132–134]
	MET−409	NASH	liver fat content↓	Humans[135]
	Cilofexor	NASH	hepatic steatosis↓ serum bile acid↓	Humans[136]
	EDP−305	NASH	liver fat content、ALT↓	Humans[137]
	Tropifexor	cholestasis	Regulates the gut-liver axis	Pigs[138]
	RDX8940	NASH	hepatic steatosis, IR[a]↓	Mice[139]
	INT767	NAFLD；Diabetes	GLP−1↑, IR[a]↓, hepatic steatosis↓ inflammation response↓	Mice[140]
Receptor Inhibitors	UDCA	NAFLD；Diabetes	IR[a], BG, fat accumulation↓ inflammation response↓	Humans[141,142]
	GUDCA	Diabetes	T-LCA, *Bacteroides fragilis*↑ TGR5, UCP−1↑	Mice[45]

serum ileal FGF-15 levels in BDL mice, which suppressed hepatic *cyp7a1* expression and de novo BA synthesis through the intestinal FXR-FGF-15 signaling pathway; moreover, it enhanced BA excretion, thereby preventing BA-induced excessive liver injury and fibrosis [130]. Another randomized controlled clinical trial evaluated the effects of *Lactobacillus acidophilus* on patients with cholestasis. Similarly, *L. acidophilus* treatment alleviated cholestatic liver injury by inhibiting hepatic BA synthesis and enhancing fecal BA excretion [131].

7.2. Targeting BA receptors: agonists/antagonists

7.2.1. Receptor agonists

OCA, a potent FXR agonist, shows remarkable potential in treating metabolic diseases by regulating the synthesis and transport of BAs. Currently, OCA is primarily used to treat primary biliary cholangitis by inhibiting BA synthesis, improving liver function, and reducing inflammation to alleviate the disease. Additionally, OCA promotes the growth of *Firmicutes* and shows significant efficacy in alleviating NASH [132]. In a multicenter clinical trial, NASH patients in the OCA treatment group showed significant improvements in liver fibrosis [133]. Phase III clinical trials further demonstrated that OCA increases insulin sensitivity, improves the severity of T2DM, and significantly reduces hepatic steatosis and fibrosis [134]. However, this drug may cause severe pruritus, and its long-term benefits and safety require further validation.

A 12-week Phase IIa clinical trial revealed that MET-409, a non-BA, structurally optimized FXR agonist, significantly reduced liver fat content. However, at an oral dose of 50 mg/day, some patients experienced transient alanine aminotransferase (ALT) elevation, along with side effects such as pruritus and increased low-density lipoprotein cholesterol [135]. Another study evaluated the safety and efficacy of the nonsteroidal agonist Cilofexor (formerly GS-9674) in 140 NASH patients. The results indicated that patients adequately tolerated the drug over the 24-week treatment period and exhibited a significant reduction in hepatic steatosis and serum BA levels. However, it is noteworthy that a small subset of patients (4 %) experienced moderate to severe pruritus [136]. EDP-305, an oral FXR agonist, is currently being developed for treating NASH. A 12-week, double-blind Phase IIa clinical trial involving 134 NASH patients assessed the efficacy, safety, and tolerability of EDP-305. The results showed that EDP-305 significantly reduced ALT levels and hepatic fat content in the liver of patients. However, $\geq$ 5 % of patients experienced adverse reactions, including pruritus, nausea, vomiting, diarrhea, headache, and dizziness, which were dose-dependent [137]. Tropifexor (NVP-LJN452), a nonsteroidal, non-BA FXR agonist, demonstrated superior efficacy compared to OCA in various preclinical NASH models [144]. Previous studies revealed that Tropifexor activates FXR and modulates the gut-liver axis, thereby alleviating gut microbiota dysbiosis, reducing inflammatory responses, and decreasing hepatic BA accumulation; this effectively inhibits cholestatic liver injury and fibrosis and highlights the potential of Tropifexor as a treatment agent for cholestasis [138].

In addition to FXR agonists, research has also been conducted on TGR5 agonists. RDX8940, a TGR5 agonist, exhibited significant efficacy in an HFD-fed mouse model, with marked improvements in insulin sensitivity and reduction of hepatic steatosis [139]. INT767, a steroidal FXR/TGR5 dual agonist, promotes fat uptake, enhances GLP-1 secretion, regulates lipid metabolism, reduces hepatic steatosis, lowers insulin resistance, and mitigates inflammation by activating these two receptors [140].

7.2.2. Receptor inhibitors

UDCA, a BA with antagonistic effects on FXR, can inhibit FXR activity, thereby influencing systemic metabolism. Currently, UDCA has been approved as the first-line treatment for primary biliary cholangitis [145]. Additionally, UDCA has shown potential efficacy in treating T2DM and NAFLD. Following UDCA supplementation, T2DM patients exhibited a significant reductions in fasting blood glucose and glycated hemoglobin levels, along with enhanced insulin sensitivity. UDCA also alleviated oxidative stress by reducing markers such as malondialdehyde and H_2O_2 and enhancing antioxidant enzyme activity, including superoxide dismutase [141]. A recent study by Stan et al. [142] found that NAFLD patients experienced a slower disease progression after UDCA treatment, characterized by enhanced BA homeostasis, reduced liver inflammation, and decreased fat accumulation; this finding particularly demonstrated the favorable therapeutic effects of UDCA treatment on patients with simple hepatic steatosis and NASH. GUDCA also inhibits intestinal FXR activity. HFD-fed mice supplemented with GUDCA exhibited elevated taurolithocholic acid levels and increased abundance of *B. fragilis*. GUDCA also activated TGR5 in adipose tissues, upregulated UCP1 expression, and promoted white adipose tissue thermogenesis, thereby alleviating hyperglycemia in HFD-fed mice [45].

8. Summary and outlook

As signaling molecules, BAs interact with the host via receptors such as FXR and TGR5, participating in the regulation of glucose and lipid metabolism, energy balance, and immune-inflammatory responses. Within the enterohepatic circulation, gut microbiota and BAs mutually regulate each other. Dysregulation of this crosstalk, leading to microbial dysbiosis and altered bile acid homeostasis, may collectively contribute to the pathogenesis of various metabolic disorders, including obesity, T2DM, NAFLD, IBD, and cholestasis. Animal studies and clinical investigations have identified several potential therapeutic targets along the gut microbiota-BA-BA receptor signaling axis. Potential therapeutic interventions include: direct BAs supplementation (e.g., UDCA), pharmacological modulation of BA transporters (e.g., ASBT inhibitors) and receptors (e.g., FXR agonists, TGR5 antagonists), gut microbiota modulation through FMT and targeted probiotic administration.

However, developing effective and safe BA signaling-based therapies requires additional fundamental and translational research to address several key challenges: (1) significant differences exist between animal models and humans in BA pool composition and gut microbiota profiles. Given the instability of results from animal studies, caution should be exercised when extrapolating mouse model data to clinical applications. Researchers should prioritize using organoid models for preclinical testing, (2) substantial interindividual variations in BA metabolism and gut microbiota necessitate further investigation into personalized treatment approaches, (3) the dynamic interplay between microbiota and BAs requires deeper mechanistic exploration to facilitate the development of targeted probiotic and prebiotic interventions, (4) while current BA-targeted therapies have shown promise, most studies have been limited to short-term efficacy assessments. Additionally, treatment-associated pruritus remains a significant clinical challenge. Therefore, comprehensive evaluation of long-term safety and efficacy profiles is urgently needed, along with the development of more selective and safer bile acid receptor agonists/antagonists, and (5) precise quantification of BA levels in humans poses another challenge for BA therapy. Although liquid chromatography-mass spectrometry (LC-MS) is widely used for bile acid profiling due to its high sensitivity (with detection limits as low as 0.1 ng/mL), specificity, and throughput, its clinical adoption is limited by high costs and technical complexity.

In summary, the significance of BAs in physiology and pathology offers new opportunities for treating metabolic diseases. Further research is essential to elucidate the underlying mechanisms of BAs in metabolic disorders, which could provide novel insights for preventing and treating these conditions.

CRediT authorship contribution statement

Min Zhao: Writing – original draft. **Huimin Yang:** Writing – original draft. **Jiafeng Zhao:** Writing – original draft. **Chang Lv:** Writing – review & editing. **Zirou Ouyang:** Writing – review & editing. **Zijun Geng:** Writing – review & editing. **zhao jianhong:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the Hebei Natural Science Foundation (No. H2022206358) and the Foundation of Hebei Provincial Department of Finance (No. ZF2025084).

Data availability

No data was used for the research described in the article.

References

[1] B. Rimal, S.L. Collins, C.E. Tanes, E.R. Rocha, M.A. Granda, S. Solanki, N. J. Hoque, E.C. Gentry, I. Koo, E.R. Reilly, F. Hao, D. Paudel, V. Singh, T. Yan, M. S. Kim, K. Bittinger, J.P. Zackular, K.W. Krausz, D. Desai, S. Amin, J.P. Coleman, Y.M. Shah, J.E. Bisanz, F.J. Gonzalez, H.J. Vanden, G.D. Wu, B.S. Zemel, P. C. Dorrestein, E.E. Weinert, A.D. Patterson, Bile salt hydrolase catalyses formation of amine-conjugated bile acids, Nature 626 (8000) (2024) 859–863, https://doi.org/10.1038/s41586-023-06990-w.

[2] C.L. Boulange, A.L. Neves, J. Chilloux, J.K. Nicholson, M. Dumas, Impact of the gut microbiota on inflammation, obesity, and metabolic disease, Genome Med 8 (1) (2016) 42, https://doi.org/10.1186/s13073-016-0303-2.

[3] R. Burcelin, gut microbiota and immune crosstalk in metabolic disease, Biol. Aujourdhui 211 (1) (2017) 1–18, https://doi.org/10.1051/jbio/2017008.

[4] A.M. Diehl, C. Day, Cause, pathogenesis, and treatment of nonalcoholic steatohepatitis, N. Engl. J. Med 377 (21) (2017) 2063–2072, https://doi.org/10.1056/NEJMra1503519.

[5] K. Dabke, G. Hendrick, S. Devkota, The gut microbiome and metabolic syndrome, J. Clin. Invest 129 (10) (2019) 4050–4057, https://doi.org/10.1172/JCI129194.

[6] A. Perino, H. Demagny, L. Velazquez-Villegas, K. Schoonjans, Molecular physiology of bile acid signaling in health, disease, and aging, Physiol. Rev. 101 (2) (2021) 683–731, https://doi.org/10.1152/physrev.00049.2019.

[7] J.C. Prost, F. Brunner, C. Bovet, C. Grob, C. Berchtold, G. Schlotterbeck, D. Kroll, C.R. Largiader, G.M. Fiedler, P. Juillerat, A uhplc-ms/ms method for the quantification of 7alpha-hydroxy-4-cholesten-3-one to assist in diagnosis of bile acid malabsorption, Clin. Mass Spectrom. 3 (2017) 1–6, https://doi.org/10.1016/j.clinms.2017.02.001.

[8] J.Y.L. Chiang, J.M. Ferrell, Bile acids as metabolic regulators and nutrient sensors, Annu Rev. Nutr. 39 (2019) 175–200, https://doi.org/10.1146/annurev-nutr-082018-124344.

[9] J.C. Poland, C.R. Flynn, Bile acids, their receptors, and the gut microbiota, Physiol. (Bethesda) 36 (4) (2021) 235–245, https://doi.org/10.1152/physiol.00028.2020.

[10] J.Y.L. Chiang, Bile acid metabolism and signaling in liver disease and therapy, Liver Res 1 (1) (2017) 3–9, https://doi.org/10.1016/j.livres.2017.05.001.

[11] J.K. Truong, A.L. Bennett, C. Klindt, A.C. Donepudi, S.R. Malla, K.J. Pachura, A. Zaufel, T. Moustafa, P.A. Dawson, S.J. Karpen, Ileal bile acid transporter inhibition in cyp2c70 ko mice ameliorates cholestatic liver injury, J. Lipid Res 63 (9) (2022) 100261, https://doi.org/10.1016/j.jlr.2022.100261.

[12] P. Tatrai, F. Erdo, P. Krajcsi, Role of hepatocyte transporters in drug-induced liver injury (dili)-in vitro testing, Pharmaceutics 15 (1) (2022), https://doi.org/10.3390/pharmaceutics15010029.

[13] M.G. Shiha, Z. Ashgar, E.M. Fraser, M. Kurien, I. Aziz, High prevalence of primary bile acid diarrhoea in patients with functional diarrhoea and irritable bowel syndrome-diarrhoea, based on rome iii and rome iv criteria, Eclinicalmedicine 25 (2020) 100465, https://doi.org/10.1016/j.eclinm.2020.100465.

[14] M.M. Thibaut, J. Gillard, A. Dolly, M. Roumain, I.A. Leclercq, N.M. Delzenne, G. G. Muccioli, L.B. Bindels, Bile acid dysregulation is intrinsically related to cachexia in tumor-bearing mice, Cancers 13 (24) (2021), https://doi.org/10.3390/cancers13246389.

[15] J.Y.L. Chiang, P. Pathak, H. Liu, A. Donepudi, J. Ferrell, S. Boehme, Intestinal farnesoid x receptor and takeda g protein couple receptor 5 signaling in metabolic regulation, Dig. Dis. 35 (3) (2017) 241–245, https://doi.org/10.1159/000450981.

[16] B. Tang, L. Tang, S. Li, S. Liu, J. He, P. Li, S. Wang, M. Yang, L. Zhang, Y. Lei, D. Tu, X. Tang, H. Hu, Q. Ouyang, X. Chen, S. Yang, Gut microbiota alters host bile acid metabolism to contribute to intrahepatic cholestasis of pregnancy, Nat. Commun. 14 (1) (2023) 1305, https://doi.org/10.1038/s41467-023-36981-4.

[17] J.B. Lynch, E.L. Gonzalez, K. Choy, K.F. Faull, T. Jewell, A. Arellano, J. Liang, K. B. Yu, J. Paramo, E.Y. Hsiao, Gut microbiota turicibacter strains differentially modify bile acids and host lipids, Nat. Commun. 14 (1) (2023) 3669, https://doi.org/10.1038/s41467-023-39403-7.

[18] H. Shen, L. Zhou, H. Zhang, Y. Yang, L. Jiang, D. Wu, H. Shu, H. Zhang, L. Xie, K. Zhou, C. Cheng, L. Yang, J. Jiang, S. Wang, Y. Han, J. Zhu, L. Xu, Z. Liu, H. Wang, S. Yin, Dietary fiber alleviates alcoholic liver injury via bacteroides acidifaciens and subsequent ammonia detoxification, Cell Host Microbe 32 (8) (2024) 1331–1346, https://doi.org/10.1016/j.chom.2024.06.008.

[19] X. Li, J. Yang, X. Zhou, C. Dai, M. Kong, L. Xie, C. Liu, Y. Liu, D. Li, X. Ma, Y. Dai, Y. Sun, Z. Jian, X. Guo, X. Lin, Y. Li, L. Sun, X. Liu, L. Jin, H. Tang, Y. Zheng, S. Hong, Ketogenic diet-induced bile acids protect against obesity through reduced calorie absorption, Nat. Metab. 6 (7) (2024) 1397–1414, https://doi.org/10.1038/s42255-024-01072-1.

[20] J.M. Ridlon, S. Devendran, J.M. Alves, H. Doden, P.G. Wolf, G.V. Pereira, L. Ly, A. Volland, H. Takei, H. Nittono, T. Murai, T. Kurosawa, G.E. Chlipala, S.J. Green, A.G. Hernandez, C.J. Fields, C.L. Wright, G. Kakiyama, I. Cann, P. Kashyap, V. Mccracken, H.R. Gaskins, The 'in vivo lifestyle' of bile acid 7α-dehydroxylating bacteria: comparative genomics, metatranscriptomic, and bile acid metabolomics analysis of a defined microbial community in gnotobiotic mice, Gut Microbes 11 (3) (2020) 381–404, https://doi.org/10.1080/19490976.2019.1618173.

[21] L. Yao, S.C. Seaton, S. Ndousse-Fetter, A.A. Adhikari, N. Dibenedetto, A.I. Mina, A.S. Banks, L. Bry, A.S. Devlin, A selective gut bacterial bile salt hydrolase alters host metabolism, Elife 7 (2018), https://doi.org/10.7554/eLife.37182.
[22] S.R. Sinha, Y. Haileselassie, L.P. Nguyen, C. Tropini, M. Wang, L.S. Becker, D. Sim, K. Jarr, E.T. Spear, G. Singh, H. Namkoong, K. Bittinger, M.A. Fischbach, J.L. Sonnenburg, A. Habtezion, Dysbiosis-induced secondary bile acid deficiency promotes intestinal inflammation, Cell Host Microbe 27 (4) (2020) 659–670, https://doi.org/10.1016/j.chom.2020.01.021.
[23] Q. Zhang, Y. Feng, L. Deng, F. Feng, L. Wang, Q. Zhou, Q. Luo, Sigb plays a major role in listeria monocytogenes tolerance to bile stress, Int. J. Food Microbiol. 145 (1) (2011) 238–243, https://doi.org/10.1016/j.ijfoodmicro.2010.12.028.
[24] M. Vital, T. Rud, S. Rath, D.H. Pieper, D. Schlüter, Diversity of bacteria exhibiting bile acid-inducible 7α-dehydroxylation genes in the human gut, Comput. Struct. Biotechnol. J. 17 (2019) 1016–1019, https://doi.org/10.1016/j.csbj.2019.07.012.
[25] J. Lee, H. Arai, Y. Nakamura, S. Fukiya, M. Wada, A. Yokota, Contribution of the 7beta-hydroxysteroid dehydrogenase from ruminococcus gnavus n53 to ursodeoxycholic acid formation in the human colon, J. Lipid Res. 54 (11) (2013) 3062–3069, https://doi.org/10.1194/jlr.M039834.
[26] Y. Sato, K. Atarashi, D.R. Plichta, Y. Arai, S. Sasajima, S.M. Kearney, W. Suda, K. Takeshita, T. Sasaki, S. Okamoto, A.N. Skelly, Y. Okamura, H. Vlamakis, Y. Li, T. Tanoue, H. Takei, H. Nittono, S. Narushima, J. Irie, H. Itoh, K. Moriya, Y. Sugiura, M. Suematsu, N. Moritoki, S. Shibata, D.R. Littman, M.A. Fischbach, Y. Uwamino, T. Inoue, A. Honda, M. Hattori, T. Murai, R.J. Xavier, N. Hirose, K. Honda, Novel bile acid biosynthetic pathways are enriched in the microbiome of centenarians, Nature 599 (7885) (2021) 458–464, https://doi.org/10.1038/s41586-021-03832-5.
[27] T. Eggert, D. Bakonyi, W. Hummel, Enzymatic routes for the synthesis of ursodeoxycholic acid, J. Biotechnol. 191 (2014) 11–21, https://doi.org/10.1016/j.jbiotec.2014.08.006.
[28] S.M. Mythen, S. Devendran, C. Méndez-García, I. Cann, J.M. Ridlon, Targeted synthesis and characterization of a gene cluster encoding nad(p)h-dependent 3α-, 3β-, and 12α-hydroxysteroid dehydrogenases from eggerthella cag:298, a gut metagenomic sequence, Appl. Environ. Microbiol 84 (7) (2018), https://doi.org/10.1128/AEM.02475-17.
[29] S.C. Harris, S. Devendran, C. Méndez-García, S.M. Mythen, C.L. Wright, C. J. Fields, A.G. Hernandez, I. Cann, P.B. Hylemon, J.M. Ridlon, Bile acid oxidation by eggerthella lenta strains c592 and dsm 2243(t), Gut Microbes 9 (6) (2018) 523–539, https://doi.org/10.1080/19490976.2018.1458180.
[30] H. Doden, L.A. Sallam, S. Devendran, L. Ly, G. Doden, S.L. Daniel, J.M.P. Alves, J. M. Ridlon, Metabolism of oxo-bile acids and characterization of recombinant 12α-hydroxysteroid dehydrogenases from bile acid 7α-dehydroxylating human gut bacteria, Appl. Environ. Microbiol 84 (10) (2018), https://doi.org/10.1128/AEM.00235-18.
[31] H.L. Doden, P.G. Wolf, H.R. Gaskins, K. Anantharaman, J.M.P. Alves, J.M. Ridlon, Completion of the gut microbial epi-bile acid pathway, Gut Microbes 13 (1) (2021) 1–20, https://doi.org/10.1080/19490976.2021.1907271.
[32] M. Bourgin, A. Kriaa, H. Mkaouar, V. Mariaule, A. Jablaoui, E. Maguin, M. Rhimi, Bile salt hydrolases: at the crossroads of microbiota and human health, Microorganisms 9 (6) (2021), https://doi.org/10.3390/microorganisms9061122.
[33] M.H. Foley, S. O'Flaherty, R. Barrangou, C.M. Theriot, Bile salt hydrolases: gatekeepers of bile acid metabolism and host-microbiome crosstalk in the gastrointestinal tract, Plos Pathog. 15 (3) (2019) e1007581, https://doi.org/10.1371/journal.ppat.1007581.
[34] S. Marion, L. Desharnais, N. Studer, Y. Dong, M.D. Notter, S. Poudel, L. Menin, A. Janowczyk, R.L. Hettich, S. Hapfelmeier, R. Bernier-Latmani, Biogeography of microbial bile acid transformations along the murine gut, J. Lipid Res 61 (11) (2020) 1450–1463, https://doi.org/10.1194/jlr.RA120001021.
[35] B. Rimal, S.L. Collins, C.E. Tanes, E.R. Rocha, M.A. Granda, S. Solanki, N. J. Hoque, E.C. Gentry, I. Koo, E.R. Reilly, F. Hao, D. Paudel, V. Singh, T. Yan, M. S. Kim, K. Bittinger, J.P. Zackular, K.W. Krausz, D. Desai, S. Amin, J.P. Coleman, Y.M. Shah, J.E. Bisanz, F.J. Gonzalez, J.P. Vanden Heuvel, G.D. Wu, B.S. Zemel, P. C. Dorrestein, E.E. Weinert, A.D. Patterson, Bile salt hydrolase catalyses formation of amine-conjugated bile acids, Nature 626 (8000) (2024) 859–863, https://doi.org/10.1038/s41586-023-06990-w.
[36] D.V. Guzior, M. Okros, M. Shivel, B. Armwald, C. Bridges, Y. Fu, C. Martin, A. L. Schilmiller, W.M. Miller, K.M. Ziegler, M.D. Sims, M.E. Maddens, S.F. Graham, R.P. Hausinger, R.A. Quinn, Bile salt hydrolase acyltransferase activity expands bile acid diversity, Nature 626 (8000) (2024) 852–858, https://doi.org/10.1038/s41586-024-07017-8.
[37] D.V. Guzior, R.A. Quinn, Review: microbial transformations of human bile acids, Microbiome 9 (1) (2021) 140, https://doi.org/10.1186/s40168-021-01101-1.
[38] M. Funabashi, T.L. Grove, M. Wang, Y. Varma, M.E. Mcfadden, L.C. Brown, C. Guo, S. Higginbottom, S.C. Almo, M.A. Fischbach, A metabolic pathway for bile acid dehydroxylation by the gut microbiome, Nature 582 (7813) (2020) 566–570, https://doi.org/10.1038/s41586-020-2396-4.
[39] S. Barman, K.L. Buzoglu, X. Yang, M. Nagarkatti, P. Nagarkatti, A.W. Decho, C. Tang, Facially amphiphilic bile acid-functionalized antimicrobials: combating pathogenic bacteria, fungi, and their biofilms, Acs Infect. Dis. 9 (9) (2023) 1769–1782, https://doi.org/10.1021/acsinfecdis.3c00266.
[40] M.C. di Gregorio, J. Cautela, L. Galantini, Physiology and physical chemistry of bile acids, Int J. Mol. Sci. 22 (4) (2021), https://doi.org/10.3390/ijms22041780.
[41] R. Thanissery, J.A. Winston, C.M. Theriot, Inhibition of spore germination, growth, and toxin activity of clinically relevant c. Difficile strains by gut microbiota derived secondary bile acids, Anaerobe 45 (2017) 86–100, https://doi.org/10.1016/j.anaerobe.2017.03.004.
[42] T.H. Sannasiddappa, P.A. Lund, S.R. Clarke, In vitro antibacterial activity of unconjugated and conjugated bile salts on Staphylococcus aureus, Front Microbiol 8 (2017) 1581, https://doi.org/10.3389/fmicb.2017.01581.
[43] Y. Peng, S. Wang, Y. Zhang, M. Chen, K. He, Q. Li, W. Huang, W. Zhang, Effects of bile acids on the growth, composition and metabolism of gut bacteria, Npj Biofilms Micro 10 (1) (2024) 112, https://doi.org/10.1038/s41522-024-00566-w.
[44] E.S. Friedman, Y. Li, T.D. Shen, J. Jiang, L. Chau, L. Adorini, F. Babakhani, J. Edwards, D. Shapiro, C. Zhao, R.M. Carr, K. Bittinger, H. Li, G.D. Wu, Fxr-dependent modulation of the human small intestinal microbiome by the bile acid derivative obeticholic acid, Gastroenterology 155 (6) (2018) 1741–1752, https://doi.org/10.1053/j.gastro.2018.08.022.
[45] B. Chen, Y. Bai, F. Tong, J. Yan, R. Zhang, Y. Zhong, H. Tan, X. Ma, Glycoursodeoxycholic acid regulates bile acids level and alters gut microbiota and glycolipid metabolism to attenuate diabetes, Gut Microbes. 15 (1) (2023) 2192155, https://doi.org/10.1080/19490976.2023.2192155.
[46] L. Zhang, Z. Zheng, H. Huang, Y. Fu, T. Chen, C. Liu, Q. Yi, C. Lin, Y. Zeng, Q. Ou, Y. Zeng, Multi-omics reveals deoxycholic acid modulates bile acid metabolism via the gut microbiota to antagonize carbon tetrachloride-induced chronic liver injury, Gut Microbes 16 (1) (2024) 2323236, https://doi.org/10.1080/19490976.2024.2323236.
[47] T. Inagaki, A. Moschetta, Y. Lee, L. Peng, G. Zhao, M. Downes, R.T. Yu, J. M. Shelton, J.A. Richardson, J.J. Repa, D.J. Mangelsdorf, S.A. Kliewer, Regulation of antibacterial defense in the small intestine by the nuclear bile acid receptor, Proc. Natl. Acad. Sci. 103 (10) (2006) 3920–3925, https://doi.org/10.1073/pnas.0509592103.
[48] H. Xu, F. Fang, K. Wu, J. Song, Y. Li, X. Lu, J. Liu, L. Zhou, W. Yu, F. Yu, J. Gao, Gut microbiota-bile acid crosstalk regulates murine lipid metabolism via the intestinal fxr-fgf19 axis in diet-induced humanized dyslipidemia, Microbiome 11 (1) (2023) 262, https://doi.org/10.1186/s40168-023-01709-5.
[49] D. Zhang, W. Lv, Y. Xu, Z. Zhang, S. Zeng, W. Zhang, L. Gong, L. Shao, M. Zhang, T. He, Y. Liu, Y. Wang, L. Liu, X. Hu, Microbial bile acid metabolite ameliorates mycophenolate mofetil-induced gastrointestinal toxicity through vitamin d3 receptor, Am. J. Transpl. 24 (7) (2024) 1132–1145, https://doi.org/10.1016/j.ajt.2024.02.029.
[50] M.M. Thibaut, L.B. Bindels, Crosstalk between bile acid-activated receptors and microbiome in entero-hepatic inflammation, Trends Mol. Med. 28 (3) (2022) 223–236, https://doi.org/10.1016/j.molmed.2021.12.006.
[51] P. Lefebvre, B. Cariou, F. Lien, F. Kuipers, B. Staels, Role of bile acids and bile acid receptors in metabolic regulation, Physiol. Rev. 89 (1) (2009) 147–191, https://doi.org/10.1152/physrev.00010.2008.
[52] J.M. Donkers, R.L.P. Roscam Abbing, S.F.J. van de Graaf, Developments in bile salt based therapies: a critical overview, Biochem Pharm. 161 (2019) 1–13, https://doi.org/10.1016/j.bcp.2018.12.018.
[53] A. Carotti, M. Marinozzi, C. Custodi, B. Cerra, R. Pellicciari, A. Gioiello, A. Macchiarulo, Beyond bile acids: targeting farnesoid x receptor (fxr) with natural and synthetic ligands, Curr. Top. Med Chem. 14 (19) (2014) 2129–2142, https://doi.org/10.2174/1568026614666141112094058.
[54] S.I. Sayin, A. Wahlstrom, J. Felin, S. Jantti, H. Marschall, K. Bamberg, B. Angelin, T. Hyotylainen, M. Oresic, F. Backhed, Gut microbiota regulates bile acid metabolism by reducing the levels of tauro-beta-muricholic acid, a naturally occurring fxr antagonist, Cell Metab. 17 (2) (2013) 225–235, https://doi.org/10.1016/j.cmet.2013.01.003.
[55] J.Y.L. Chiang, Negative feedback regulation of bile acid metabolism: impact on liver metabolism and diseases, Hepatology 62 (4) (2015) 1315–1317, https://doi.org/10.1002/hep.27964.
[56] J.Y.L. Chiang, J.M. Ferrell, Discovery of farnesoid x receptor and its role in bile acid metabolism, Mol. Cell Endocrinol. 548 (2022) 111618, https://doi.org/10.1016/j.mce.2022.111618.
[57] J. Cai, B. Rimal, C. Jiang, J.Y.L. Chiang, A.D. Patterson, Bile acid metabolism and signaling, the microbiota, and metabolic disease, Pharm. Ther. 237 (2022) 108238, https://doi.org/10.1016/j.pharmthera.2022.108238.
[58] M.M. Holter, M.K. Chirikjian, V.N. Govani, B.P. Cummings, Tgr5 signaling in hepatic metabolic health, Nutrients 12 (9) (2020), https://doi.org/10.3390/nu12092598.
[59] R.E. Kuhre, N.J. Wewer Albrechtsen, O. Larsen, S.L. Jepsen, E. Balk-Moller, D. B. Andersen, C.F. Deacon, K. Schoonjans, F. Reimann, F.M. Gribble, R. Albrechtsen, B. Hartmann, M.M. Rosenkilde, J.J. Holst, Bile acids are important direct and indirect regulators of the secretion of appetite- and metabolism-regulating hormones from the gut and pancreas, Mol. Metab. 11 (2018) 84–95, https://doi.org/10.1016/j.molmet.2018.03.007.
[60] P. Pathak, H. Liu, S. Boehme, C. Xie, K.W. Krausz, F. Gonzalez, J.Y.L. Chiang, Farnesoid x receptor induces takeda g-protein receptor 5 cross-talk to regulate bile acid synthesis and hepatic metabolism, J. Biol. Chem. 292 (26) (2017) 11055–11069, https://doi.org/10.1074/jbc.M117.784322.
[61] A.C. Donepudi, S. Boehme, F. Li, J.Y.L. Chiang, G-protein-coupled bile acid receptor plays a key role in bile acid metabolism and fasting-induced hepatic steatosis in mice, Hepatology 65 (3) (2017) 813–827, https://doi.org/10.1002/hep.28707.
[62] X. Li, L. Feng, Q. Kuang, X. Wang, J. Yang, X. Niu, L. Gao, L. Huang, P. Luo, L. Li, Microplastics cause hepatotoxicity in diabetic mice by disrupting glucolipid metabolism via pp2a/ampk/hnf4a and promoting fibrosis via the wnt/beta-catenin pathway, Environ. Toxicol. 39 (2) (2024) 1018–1030, https://doi.org/10.1002/tox.24034.
[63] H. Yan, W. Yang, F. Zhou, X. Li, Q. Pan, Z. Shen, G. Han, A. Newell-Fugate, Y. Tian, R. Majeti, W. Liu, Y. Xu, C. Wu, K. Allred, C. Allred, Y. Sun, S. Guo,

Estrogen improves insulin sensitivity and suppresses gluconeogenesis via the transcription factor foxo1, Diabetes 68 (2) (2019) 291–304, https://doi.org/10.2337/db18-0638.
[64] W.H. Park, Y.K. Pak, Insulin-dependent suppression of cholesterol 7alpha-hydroxylase is a possible link between glucose and cholesterol metabolisms, Exp. Mol. Med 43 (10) (2011) 571–579, https://doi.org/10.3858/emm.2011.43.10.064.
[65] K. Yamagata, H. Daitoku, Y. Shimamoto, H. Matsuzaki, K. Hirota, J. Ishida, A. Fukamizu, Bile acids regulate gluconeogenic gene expression via small heterodimer partner-mediated repression of hepatocyte nuclear factor 4 and foxo1, J. Biol. Chem. 279 (22) (2004) 23158–23165, https://doi.org/10.1074/jbc.M314322200.
[66] K. Ma, P.K. Saha, L. Chan, Farnesoid x receptor is essential for normal glucose homeostasis, J. Clin. Invest 116 (4) (2006) 1102–1109, https://doi.org/10.1172/JCI25604.
[67] T. Zhao, J. Wang, A. He, S. Wang, Y. Chen, J. Lu, J. Lv, S. Li, J. Wang, M. Qian, H. Li, X. Shen, Mebhydrolin ameliorates glucose homeostasis in type 2 diabetic mice by functioning as a selective fxr antagonist, Metabolism 119 (2021) 154771, https://doi.org/10.1016/j.metabol.2021.154771.
[68] L. Li, H. Zhao, B. Chen, Z. Fan, N. Li, J. Yue, Q. Ye, Fxr activation alleviates tacrolimus-induced post-transplant diabetes mellitus by regulating renal gluconeogenesis and glucose uptake, J. Transl. Med 17 (1) (2019) 418, https://doi.org/10.1186/s12967-019-02170-5.
[69] G.M. Soares, S.L. Balbo, G.A. Bronczek, J.F. Vettorazzi, C. Marmentini, L. Zangerolamo, L.A. Velloso, E.M. Carneiro, Vertical sleeve gastrectomy improves glucose-insulin homeostasis by enhancing beta-cell function and survival via fgf15/19, Am. J. Physiol. Endocrinol. Metab. 326 (2) (2024) E134–E147, https://doi.org/10.1152/ajpendo.00218.2023.
[70] Q. Wang, H. Lin, C. Shen, M. Zhang, X. Wang, M. Yuan, M. Yuan, S. Jia, Z. Cao, C. Wu, B. Chen, A. Gao, Y. Bi, G. Ning, W. Wang, J. Wang, R. Liu, Gut microbiota regulates postprandial glp-1 response via ileal bile acid-tgr5 signaling, Gut Microbes 15 (2) (2023) 2274124, https://doi.org/10.1080/19490976.2023.2274124.
[71] J. She, G. Tuerhongjiang, M. Guo, J. Liu, X. Hao, L. Guo, N. Liu, W. Xi, T. Zheng, B. Du, B. Lou, X. Gao, X. Yuan, Y. Yu, Y. Zhang, F. Gao, X. Zhuo, Y. Xiong, X. Zhang, J. Yu, Z. Yuan, Y. Wu, Statins aggravate insulin resistance through reduced blood glucagon-like peptide-1 levels in a microbiota-dependent manner, Cell Metab. 36 (2) (2024) 408–421, https://doi.org/10.1016/j.cmet.2023.12.027.
[72] D.J. Drucker, Glp-1 physiology informs the pharmacotherapy of obesity, Mol. Metab. 57 (2022) 101351, https://doi.org/10.1016/j.molmet.2021.101351.
[73] X. Song, X. Sun, S.F. Oh, M. Wu, Y. Zhang, W. Zheng, N. Geva-Zatorsky, R. Jupp, D. Mathis, C. Benoist, D.L. Kasper, Microbial bile acid metabolites modulate gut rorgamma(+) regulatory t cell homeostasis, Nature 577 (7790) (2020) 410–415, https://doi.org/10.1038/s41586-019-1865-0.
[74] J. Xiao, L. Dong, S. Liu, F. Meng, C. Xie, X. Lu, W.J. Zhang, J. Luo, B. Song, Bile acids-mediated intracellular cholesterol transport promotes intestinal cholesterol absorption and npc1l1 recycling, Nat. Commun. 14 (1) (2023) 6469, https://doi.org/10.1038/s41467-023-42179-5.
[75] S. Farr, B. Stankovic, S. Hoffman, H. Masoudpoor, C. Baker, J. Taher, A.E. Dean, S. Anakk, K. Adeli, Bile acid treatment and fxr agonism lower postprandial lipemia in mice, Am. J. Physiol. Gastrointest. Liver Physiol. 318 (4) (2020) G682–G693, https://doi.org/10.1152/ajpgi.00386.2018.
[76] A. Peng, S. Liu, L. Fang, Z. Zhu, Y. Zhou, S. Yue, Z. Ma, X. Liu, S. Xue, Y. Qiu, R. Qi, Inonotus obliquus and its bioactive compounds alleviate non-alcoholic fatty liver disease via regulating fxr/shp/srebp-1c axis, Eur. J. Pharm. 921 (2022) 174841, https://doi.org/10.1016/j.ejphar.2022.174841.
[77] X. Duan, Q. Meng, C. Wang, Z. Liu, Q. Liu, H. Sun, P. Sun, X. Yang, X. Huo, J. Peng, K. Liu, Calycosin attenuates triglyceride accumulation and hepatic fibrosis in murine model of non-alcoholic steatohepatitis via activating farnesoid x receptor, Phytomedicine 25 (2017) 83–92, https://doi.org/10.1016/j.phymed.2016.12.006.
[78] M.C. Cave, H.B. Clair, J.E. Hardesty, K.C. Falkner, W. Feng, B.J. Clark, J. Sidey, H. Shi, B.A. Aqel, C.J. Mcclain, R.A. Prough, Nuclear receptors and nonalcoholic fatty liver disease, Biochim Biophys. Acta 1859 (9) (2016) 1083–1099, https://doi.org/10.1016/j.bbagrm.2016.03.002.
[79] A.M. Anisfeld, H.R. Kast-Woelbern, M.E. Meyer, S.A. Jones, Y. Zhang, K. J. Williams, T. Willson, P.A. Edwards, Syndecan-1 expression is regulated in an isoform-specific manner by the farnesoid-x receptor, J. Biol. Chem. 278 (22) (2003) 20420–20428, https://doi.org/10.1074/jbc.M302505200.
[80] H.R. Kast, C.M. Nguyen, C.J. Sinal, S.A. Jones, B.A. Laffitte, K. Reue, F. J. Gonzalez, T.M. Willson, P.A. Edwards, Farnesoid x-activated receptor induces apolipoprotein c-ii transcription: a molecular mechanism linking plasma triglyceride levels to bile acids, Mol. Endocrinol. 15 (10) (2001) 1720–1728, https://doi.org/10.1210/mend.15.10.0712.
[81] X. Prieur, H. Coste, J.C. Rodriguez, The human apolipoprotein av gene is regulated by peroxisome proliferator-activated receptor-alpha and contains a novel farnesoid x-activated receptor response element, J. Biol. Chem. 278 (28) (2003) 25468–25480, https://doi.org/10.1074/jbc.M301302200.
[82] H. Hirokane, M. Nakahara, S. Tachibana, M. Shimizu, R. Sato, Bile acid reduces the secretion of very low density lipoprotein by repressing microsomal triglyceride transfer protein gene expression mediated by hepatocyte nuclear factor-4, J. Biol. Chem. 279 (44) (2004) 45685–45692, https://doi.org/10.1074/jbc.M404255200.
[83] T. Claudel, Y. Inoue, O. Barbier, D. Duran-Sandoval, V. Kosykh, J. Fruchart, J. Fruchart, F.J. Gonzalez, B. Staels, Farnesoid x receptor agonists suppress hepatic apolipoprotein ciii expression, Gastroenterology 125 (2) (2003) 544–555, https://doi.org/10.1016/s0016-5085(03)00896-5.
[84] M. Watanabe, S.M. Houten, C. Mataki, M.A. Christoffolete, B.W. Kim, H. Sato, N. Messaddeq, J.W. Harney, O. Ezaki, T. Kodama, K. Schoonjans, A.C. Bianco, J. Auwerx, Bile acids induce energy expenditure by promoting intracellular thyroid hormone activation, Nature 439 (7075) (2006) 484–489, https://doi.org/10.1038/nature04330.
[85] S. Fang, J.M. Suh, S.M. Reilly, E. Yu, O. Osborn, D. Lackey, E. Yoshihara, A. Perino, S. Jacinto, Y. Lukasheva, A.R. Atkins, A. Khvat, B. Schnabl, R.T. Yu, D. A. Brenner, S. Coulter, C. Liddle, K. Schoonjans, J.M. Olefsky, A.R. Saltiel, M. Downes, R.M. Evans, Intestinal fxr agonism promotes adipose tissue browning and reduces obesity and insulin resistance, Nat. Med 21 (2) (2015) 159–165, https://doi.org/10.1038/nm.3760.
[86] J. Yang, H.D. de Vries, A. Mayeuf-Louchart, J.H. Stroeve, V.W. Bloks, M. Koehorst, H. Duez, B. Staels, F. Kuipers, T. van Zutphen, Role of bile acid receptor fxr in development and function of brown adipose tissue, Biochim Biophys. Acta Mol. Cell Biol. Lipids 1868 (2) (2023) 159257, https://doi.org/10.1016/j.bbalip.2022.159257.
[87] S.L. Collins, J.G. Stine, J.E. Bisanz, C.D. Okafor, A.D. Patterson, Bile acids and the gut microbiota: metabolic interactions and impacts on disease, Nat. Rev. Microbiol 21 (4) (2023) 236–247, https://doi.org/10.1038/s41579-022-00805-x.
[88] H. Sun, Y. Guo, H. Wang, A. Yin, J. Hu, T. Yuan, S. Zhou, W. Xu, P. Wei, S. Yin, P. Liu, X. Guo, Y. Tang, Y. Yan, Z. Luo, M. Wang, Q. Liang, P. Wu, A. Zhang, Z. Zhou, Y. Chen, Y. Li, J. Li, J. Shan, W. Zhou, Gut commensal parabacteroides distasonis alleviates inflammatory arthritis, Gut 72 (9) (2023) 1664–1677, https://doi.org/10.1136/gutjnl-2022-327756.
[89] S. Hang, D. Paik, L. Yao, E. Kim, J. Trinath, J. Lu, S. Ha, B.N. Nelson, S.P. Kelly, L. Wu, Y. Zheng, R.S. Longman, F. Rastinejad, A.S. Devlin, M.R. Krout, M. A. Fischbach, D.R. Littman, J.R. Huh, Bile acid metabolites control t(h)17 and t (reg) cell differentiation, Nature 576 (7785) (2019) 143–148, https://doi.org/10.1038/s41586-019-1785-z.
[90] W. Gui, M.J. Hole, A. Molinaro, K. Edlund, K.K. Jorgensen, H. Su, B. Begher-Tibbe, N. Gassler, C.V. Schneider, U. Muthukumarasamy, A. Mohs, L. Liao, J. Jaeger, C.J. Mertens, I. Bergheim, T. Strowig, J.G. Hengstler, J.R. Hov, H. Marschall, C. Trautwein, K.M. Schneider, Colitis ameliorates cholestatic liver disease via suppression of bile acid synthesis, Nat. Commun. 14 (1) (2023) 3304, https://doi.org/10.1038/s41467-023-38840-8.
[91] B. Guan, J. Tong, H. Hao, Z. Yang, K. Chen, H. Xu, A. Wang, Bile acid coordinates microbiota homeostasis and systemic immunometabolism in cardiometabolic diseases, Acta Pharm. Sin. B 12 (5) (2022) 2129–2149, https://doi.org/10.1016/j.apsb.2021.12.011.
[92] Y. Wang, W. Chen, D. Yu, B.M. Forman, W. Huang, The g-protein-coupled bile acid receptor, gpbar1 (tgr5), negatively regulates hepatic inflammatory response through antagonizing nuclear factor kappa light-chain enhancer of activated b cells (nf-kappab) in mice, Hepatology 54 (4) (2011) 1421–1432, https://doi.org/10.1002/hep.24525.
[93] O. Chavez-Talavera, A. Tailleux, P. Lefebvre, B. Staels, Bile acid control of metabolism and inflammation in obesity, type 2 diabetes, dyslipidemia, and nonalcoholic fatty liver disease, Gastroenterology 152 (7) (2017) 1679–1694, https://doi.org/10.1053/j.gastro.2017.01.055.
[94] D.E. Lackey, J.M. Olefsky, Regulation of metabolism by the innate immune system, Nat. Rev. Endocrinol. 12 (1) (2016) 15–28, https://doi.org/10.1038/nrendo.2015.189.
[95] Global, regional, and national burden of diabetes from 1990 to 2021, with projections of prevalence to 2050: a systematic analysis for the global burden of disease study 2021, Lancet. 402(10397) (2023) 203-234, https://doi.org/10.1016/S0140-6736(23)01301-6.
[96] R.A. Haeusler, B. Astiarraga, S. Camastra, D. Accili, E. Ferrannini, Human insulin resistance is associated with increased plasma levels of 12alpha-hydroxylated bile acids, Diabetes 62 (12) (2013) 4184–4191, https://doi.org/10.2337/db13-0639.
[97] U. Axling, M. Cavalera, E. Degerman, M. Gafvels, G. Eggertsen, C. Holm, Increased whole body energy expenditure and protection against diet-induced obesity in cyp8b1-deficient mice is accompanied by altered adipose tissue features, Adipocyte 9 (1) (2020) 587–599, https://doi.org/10.1080/21623945.2020.1827519.
[98] P. Pathak, J.Y.L. Chiang, Sterol 12alpha-hydroxylase aggravates dyslipidemia by activating the ceramide/mtorc1/srebp-1c pathway via fgf21 and fgf15, Gene Expr. 19 (3) (2019) 161–173, https://doi.org/10.3727/105221619X15529371970455.
[99] Y. Liu, J. Tu, L. Shi, Z. Fang, M. Fan, J. Zhang, L. Ding, Y. Chen, Y. Wang, E. Zhang, S. Xu, N. Sharma, J.D. Gillece, L.J. Reining, L. Jin, W. Huang, Cyp8b1 downregulation mediates the metabolic effects of vertical sleeve gastrectomy in mice, Hepatology 79 (5) (2024) 1005–1018, https://doi.org/10.1097/HEP.0000000000000627.
[100] M. Wei, F. Huang, L. Zhao, Y. Zhang, W. Yang, S. Wang, M. Li, X. Han, K. Ge, C. Qu, C. Rajani, G. Xie, X. Zheng, A. Zhao, Z. Bian, W. Jia, A dysregulated bile acid-gut microbiota axis contributes to obesity susceptibility, Ebiomedicine 55 (2020) 102766, https://doi.org/10.1016/j.ebiom.2020.102766.
[101] W. Gou, C. Ling, Y. He, Z. Jiang, Y. Fu, F. Xu, Z. Miao, T. Sun, J. Lin, H. Zhu, H. Zhou, Y. Chen, J. Zheng, Interpretable machine learning framework reveals robust gut microbiome features associated with type 2 diabetes, Diabetes Care 44 (2) (2021) 358–366, https://doi.org/10.2337/dc20-1536.
[102] M. Gurung, Z. Li, H. You, R. Rodrigues, D.B. Jump, A. Morgun, N. Shulzhenko, Role of gut microbiota in type 2 diabetes pathophysiology, Ebiomedicine 51 (2020) 102590, https://doi.org/10.1016/j.ebiom.2019.11.051.

[103] A.S. Devlin, M.A. Fischbach, A biosynthetic pathway for a prominent class of microbiota-derived bile acids, Nat. Chem. Biol. 11 (9) (2015) 685–690, https://doi.org/10.1038/nchembio.1864.
[104] G. Feng, L. Valenti, V.W. Wong, Y.M. Fouad, Y. Yilmaz, W. Kim, G. Sebastiani, Z. M. Younossi, V. Hernandez-Gea, M. Zheng, Recompensation in cirrhosis: unravelling the evolving natural history of nonalcoholic fatty liver disease, Nat. Rev. Gastroenterol. Hepatol. 21 (1) (2024) 46–56, https://doi.org/10.1038/s41575-023-00846-4.
[105] N. Stefan, K. Cusi, A global view of the interplay between non-alcoholic fatty liver disease and diabetes, Lancet Diabetes Endocrinol. 10 (4) (2022) 284–296, https://doi.org/10.1016/S2213-8587(22)00003-1.
[106] M. Mouzaki, A.Y. Wang, R. Bandsma, E.M. Comelli, B.M. Arendt, L. Zhang, S. Fung, S.E. Fischer, I.G. Mcgilvray, J.P. Allard, Bile acids and dysbiosis in non-alcoholic fatty liver disease, Plos One 11 (5) (2016) e151829, https://doi.org/10.1371/journal.pone.0151829.
[107] C. Caussy, C. Hsu, S. Singh, S. Bassirian, J. Kolar, C. Faulkner, N. Sinha, R. Bettencourt, N. Gara, M.A. Valasek, B. Schnabl, L. Richards, D.A. Brenner, A. F. Hofmann, R. Loomba, Serum bile acid patterns are associated with the presence of nafld in twins, and dose-dependent changes with increase in fibrosis stage in patients with biopsy-proven nafld, Aliment Pharm. Ther. 49 (2) (2019) 183–193, https://doi.org/10.1111/apt.15035.
[108] J. Kuang, J. Wang, Y. Li, M. Li, M. Zhao, K. Ge, D. Zheng, K.C.P. Cheung, B. Liao, S. Wang, T. Chen, Y. Zhang, C. Wang, G. Ji, P. Chen, H. Zhou, C. Xie, A. Zhao, W. Jia, X. Zheng, W. Jia, Hyodeoxycholic acid alleviates non-alcoholic fatty liver disease through modulating the gut-liver axis, Cell Metab. 35 (10) (2023) 1752–1766, https://doi.org/10.1016/j.cmet.2023.07.011.
[109] L.E. Armstrong, G.L. Guo, Role of fxr in liver inflammation during nonalcoholic steatohepatitis, Curr. Pharm. Rep. 3 (2) (2017) 92–100, https://doi.org/10.1007/s40495-017-0085-2.
[110] B.L. Clifford, L.R. Sedgeman, K.J. Williams, P. Morand, A. Cheng, K.E. Jarrett, A. P. Chan, M.C. Brearley-Sholto, A. Wahlström, J.W. Ashby, W. Barshop, J. Wohlschlegel, A.C. Calkin, Y. Liu, A. Thorell, P.J. Meikle, B.G. Drew, J.J. Mack, H. Marschall, E.J. Tarling, P.A. Edwards, T.Q. de Aguiar Vallim, Fxr activation protects against nafld via bile-acid-dependent reductions in lipid absorption, Cell Metab. 33 (8) (2021) 1671–1684, https://doi.org/10.1016/j.cmet.2021.06.012.
[111] A.B. Singh, B. Dong, F.B. Kraemer, Y. Xu, Y. Zhang, J. Liu, Farnesoid x receptor activation by obeticholic acid elevates liver low-density lipoprotein receptor expression by mrna stabilization and reduces plasma low-density lipoprotein cholesterol in mice, Arterioscler. Thromb. Vasc. Biol. 38 (10) (2018) 2448–2459, https://doi.org/10.1161/ATVBAHA.118.311122.
[112] A. Carino, S. Cipriani, S. Marchianò, M. Biagioli, C. Santorelli, A. Donini, A. Zampella, M.C. Monti, S. Fiorucci, Bar502, a dual fxr and gpbar1 agonist, promotes browning of white adipose tissue and reverses liver steatosis and fibrosis, Sci. Rep. 7 (2017) 42801, https://doi.org/10.1038/srep42801.
[113] H. Duboc, S. Rajca, D. Rainteau, D. Benarous, M. Maubert, E. Quervain, G. Thomas, V. Barbu, L. Humbert, G. Despras, C. Bridonneau, F. Dumetz, J. Grill, J. Masliah, L. Beaugerie, J. Cosnes, O. Chazouillères, R. Poupon, C. Wolf, J. Mallet, P. Langella, G. Trugnan, H. Sokol, P. Seksik, Connecting dysbiosis, bile-acid dysmetabolism and gut inflammation in inflammatory bowel diseases, Gut 62 (4) (2013) 531–539, https://doi.org/10.1136/gutjnl-2012-302578.
[114] L. Chen, T. Jiao, W. Liu, Y. Luo, J. Wang, X. Guo, X. Tong, Z. Lin, C. Sun, K. Wang, Y. He, Y. Zhang, H. Xu, J. Wang, J. Zuo, Q. Ding, S. He, F.J. Gonzalez, C. Xie, Hepatic cytochrome p450 8b1 and cholic acid potentiate intestinal epithelial injury in colitis by suppressing intestinal stem cell renewal, Cell Stem Cell 29 (9) (2022) 1366–1381, https://doi.org/10.1016/j.stem.2022.08.008.
[115] M. Zheng, Y. Zhai, Y. Yu, J. Shen, S. Chu, E. Focaccia, W. Tian, S. Wang, X. Liu, X. Yuan, Y. Wang, L. Li, B. Feng, Z. Li, X. Guo, J. Qiu, C. Zhang, J. Hou, Y. Sun, X. Yang, X. Zuo, M. Heikenwalder, Y. Li, D. Yuan, S. Li, Tnf compromises intestinal bile-acid tolerance dictating colitis progression and limited infliximab response, Cell Metab. 36 (9) (2024) 2086–2103, https://doi.org/10.1016/j.cmet.2024.06.008.
[116] Y. Ma, H. Yang, X. Wang, Y. Huang, Y. Li, G. Pan, Bile acids as signaling molecules in inflammatory bowel disease: implications for treatment strategies, J. Ethnopharmacol. 337 (Pt 3) (2025) 118968, https://doi.org/10.1016/j.jep.2024.118968.
[117] S. Rotondo-Trivette, B. Wang, C. Gayer, R. Parsana, Y. Luan, F. Sun, S. Michail, Decreased secondary faecal bile acids in children with ulcerative colitis and clostridioides difficile infection, Aliment Pharm. Ther. 54 (6) (2021) 792–804, https://doi.org/10.1111/apt.16496.
[118] L. Li, T. Liu, Y. Gu, X. Wang, R. Xie, Y. Sun, B. Wang, H. Cao, Regulation of gut microbiota-bile acids axis by probiotics in inflammatory bowel disease, Front Immunol. 13 (2022) 974305, https://doi.org/10.3389/fimmu.2022.974305.
[119] A. Wilson, A. Almousa, W.A. Teft, R.B. Kim, Attenuation of bile acid-mediated fxr and pxr activation in patients with crohn's disease, Sci. Rep. 10 (1) (2020) 1866, https://doi.org/10.1038/s41598-020-58644-w.
[120] R.M. Gadaleta, K.J. van Erpecum, B. Oldenburg, E.C.L. Willemsen, W. Renooij, S. Murzilli, L.W.J. Klomp, P.D. Siersema, M.E.I. Schipper, S. Danese, G. Penna, G. Laverny, L. Adorini, A. Moschetta, S.W.C. van Mil, Farnesoid x receptor activation inhibits inflammation and preserves the intestinal barrier in inflammatory bowel disease, Gut 60 (4) (2011) 463–472, https://doi.org/10.1136/gut.2010.212159.
[121] L. Zhang, Q. Pan, L. Zhang, H. Xia, J. Liao, X. Zhang, N. Zhao, Q. Xie, M. Liao, Y. Tan, Q. Li, J. Zhu, L. Li, S. Fan, J. Li, C. Zhang, S. Cai, J.L. Boyer, J. Chai, Runt-related transcription factor-1 ameliorates bile acid-induced hepatic inflammation in cholestasis through jak/stat3 signaling, Hepatology 77 (6) (2023) 1866–1881, https://doi.org/10.1097/HEP.0000000000000041.
[122] J.P. Arab, D. Cabrera, M. Arrese, Bile acids in cholestasis and its treatment, Ann. Hepatol. 16 (2017), https://doi.org/10.5604/01.3001.0010.5497.
[123] L. Song, Y. Hou, D. Xu, X. Dai, J. Luo, Y. Liu, Z. Huang, M. Yang, J. Chen, Y. Hu, C. Chen, Y. Tang, Z. Rao, J. Ma, M. Zheng, K. Shi, C. Cai, M. Lu, R. Tang, X. Ma, C. Xie, Y. Luo, X. Li, Z. Huang, Hepatic fxr-fgf4 is required for bile acid homeostasis via an fgfr4-lrh-1 signal node under cholestatic stress, Cell Metab. 37 (1) (2025) 104–120, https://doi.org/10.1016/j.cmet.2024.09.008.
[124] X. Yang, Y. Xu, J. Li, X. Ran, Z. Gu, L. Song, L. Zhang, L. Wen, G. Ji, R. Wang, Bile acid-gut microbiota imbalance in cholestasis and its long-term effect in mice, Msystems 9 (7) (2024) e12724, https://doi.org/10.1128/msystems.00127-24.
[125] R. Tang, Y. Wei, Y. Li, W. Chen, H. Chen, Q. Wang, F. Yang, Q. Miao, X. Xiao, H. Zhang, M. Lian, X. Jiang, J. Zhang, Q. Cao, Z. Fan, M. Wu, D. Qiu, J. Fang, A. Ansari, M.E. Gershwin, X. Ma, Gut microbial profile is altered in primary biliary cholangitis and partially restored after udca therapy, Gut 67 (3) (2018) 534–541, https://doi.org/10.1136/gutjnl-2016-313332.
[126] B. Jiang, G. Yuan, J. Wu, Q. Wu, L. Li, P. Jiang, Prevotella copri ameliorates cholestasis and liver fibrosis in primary sclerosing cholangitis by enhancing the fxr signalling pathway, Biochim Biophys. Acta Mol. Basis Dis. 1868 (3) (2022) 166320, https://doi.org/10.1016/j.bbadis.2021.166320.
[127] J.R. Allegretti, Z. Kassam, B.H. Mullish, A. Chiang, M. Carrellas, J. Hurtado, J. R. Marchesi, J.A.K. Mcdonald, A. Pechlivanis, G.F. Barker, J. Miguéns Blanco, I. Garcia-Perez, W.F. Wong, Y. Gerardin, M. Silverstein, K. Kennedy, C. Thompson, Effects of fecal microbiota transplantation with oral capsules in obese patients, Clin. Gastroenterol. Hepatol. 18 (4) (2020) 855–863, https://doi.org/10.1016/j.cgh.2019.07.006.
[128] J.R. Allegretti, Z. Kassam, J. Hurtado, J.R. Marchesi, B.H. Mullish, A. Chiang, C. C. Thompson, B.P. Cummings, Impact of fecal microbiota transplantation with capsules on the prevention of metabolic syndrome among patients with obesity, Hormones 20 (1) (2021) 209–211, https://doi.org/10.1007/s42000-020-00265-z.
[129] Z. Zhang, H. Zhou, X. Zhou, J. Sun, X. Liang, Y. Lv, L. Bai, J. Zhang, P. Gong, T. Liu, H. Yi, J. Wang, L. Zhang, Lactobacillus casei yrl577 ameliorates markers of non-alcoholic fatty liver and alters expression of genes within the intestinal bile acid pathway, Br. J. Nutr. 125 (5) (2021) 521–529, https://doi.org/10.1017/S0007114520003001.
[130] Y. Liu, K. Chen, F. Li, Z. Gu, Q. Liu, L. He, T. Shao, Q. Song, F. Zhu, L. Zhang, M. Jiang, Y. Zhou, S. Barve, X. Zhang, C.J. Mcclain, W. Feng, Probiotic lactobacillus rhamnosus gg prevents liver fibrosis through inhibiting hepatic bile acid synthesis and enhancing bile acid excretion in mice, Hepatology 71 (6) (2020) 2050–2066, https://doi.org/10.1002/hep.30975.
[131] L. Wu, J. Zhou, A. Zhou, Y. Lei, L. Tang, S. Hu, S. Wang, X. Xiao, Q. Chen, D. Tu, C. Lu, Y. Lai, Y. Li, X. Zhang, B. Tang, S. Yang, Lactobacillus acidophilus ameliorates cholestatic liver injury through inhibiting bile acid synthesis and promoting bile acid excretion, Gut Microbes 16 (1) (2024) 2390176, https://doi.org/10.1080/19490976.2024.2390176.
[132] J.M. Ferrell, J.Y.L. Chiang, Understanding bile acid signaling in diabetes: from pathophysiology to therapeutic targets, Diabetes Metab. J. 43 (3) (2019) 257–272, https://doi.org/10.4093/dmj.2019.0043.
[133] B.A. Neuschwander-Tetri, R. Loomba, A.J. Sanyal, J.E. Lavine, M.L. Van Natta, M. F. Abdelmalek, N. Chalasani, S. Dasarathy, A.M. Diehl, B. Hameed, K.V. Kowdley, A. Mccullough, N. Terrault, J.M. Clark, J. Tonascia, E.M. Brunt, D.E. Kleiner, E. Doo, Farnesoid x nuclear receptor ligand obeticholic acid for non-cirrhotic, non-alcoholic steatohepatitis (flint): a multicentre, randomised, placebo-controlled trial, Lancet 385 (9972) (2015) 956–965, https://doi.org/10.1016/S0140-6736(14)61933-4.
[134] Z.M. Younossi, V. Ratziu, R. Loomba, M. Rinella, Q.M. Anstee, Z. Goodman, P. Bedossa, A. Geier, S. Beckebaum, P.N. Newsome, D. Sheridan, M.Y. Sheikh, J. Trotter, W. Knapple, E. Lawitz, M.F. Abdelmalek, K.V. Kowdley, A.J. Montano-Loza, J. Boursier, P. Mathurin, E. Bugianesi, G. Mazzella, A. Olveira, H. Cortez-Pinto, I. Graupera, D. Orr, L.L. Gluud, J. Dufour, D. Shapiro, J. Campagna, L. Zaru, L. Macconell, R. Shringarpure, S. Harrison, A.J. Sanyal, Obeticholic acid for the treatment of non-alcoholic steatohepatitis: interim analysis from a multicentre, randomised, placebo-controlled phase 3 trial, Lancet 394 (10215) (2019) 2184–2196, https://doi.org/10.1016/S0140-6736(19)33041-7.
[135] S.A. Harrison, M.R. Bashir, K. Lee, J. Shim-Lopez, J. Lee, B. Wagner, N.D. Smith, H.C. Chen, E.J. Lawitz, A structurally optimized fxr agonist, met409, reduced liver fat content over 12 weeks in patients with non-alcoholic steatohepatitis, J. Hepatol. 75 (1) (2021) 25–33, https://doi.org/10.1016/j.jhep.2021.01.047.
[136] K. Patel, S.A. Harrison, M. Elkhashab, J.F. Trotter, R. Herring, S.E. Rojter, Z. Kayali, V.W. Wong, S. Greenbloom, S. Jayakumar, M.L. Shiffman, B. Freilich, E. J. Lawitz, E.J. Gane, E. Harting, J. Xu, A.N. Billin, C. Chung, C.S. Djedjos, G. M. Subramanian, R.P. Myers, M.S. Middleton, M. Rinella, M. Noureddin, Cilofexor, a nonsteroidal fxr agonist, in patients with noncirrhotic nash: a phase 2 randomized controlled trial, Hepatology 72 (1) (2020) 58–71, https://doi.org/10.1002/hep.31205.
[137] V. Ratziu, M.E. Rinella, B.A. Neuschwander-Tetri, E. Lawitz, D. Denham, Z. Kayali, A. Sheikh, K.V. Kowdley, T. Desta, M. Elkhashab, J. Degrauw, B. Goodwin, A. Ahmad, N. Adda, Edp-305 in patients with nash: a phase ii double-blind placebo-controlled dose-ranging study, J. Hepatol. 76 (3) (2022) 506–517, https://doi.org/10.1016/j.jhep.2021.10.018.
[138] Y. Xiao, Y. Wang, Y. Liu, W. Wang, X. Tian, S. Chen, Y. Lu, J. Du, W. Cai, A nonbile acid farnesoid x receptor agonist tropifexor potently inhibits cholestatic liver injury and fibrosis by modulating the gut-liver axis, Liver Int 41 (9) (2021) 2117–2131, https://doi.org/10.1111/liv.14906.
[139] P.D. Finn, D. Rodriguez, J. Kohler, Z. Jiang, S. Wan, E. Blanco, A.J. King, T. Chen, N. Bell, D. Dragoli, J.W. Jacobs, R. Jain, M. Leadbetter, M. Siegel, C.W. Carreras,

S. Koo-Mccoy, K. Shaw, C. Le, S. Vanegas, I. Hsu, K. Kozuka, K. Okamoto, J. S. Caldwell, J.G. Lewis, Intestinal tgr5 agonism improves hepatic steatosis and insulin sensitivity in western diet-fed mice, Am. J. Physiol. Gastrointest. Liver Physiol. 316 (3) (2019) G412–G424, https://doi.org/10.1152/ajpgi.00300.2018.

[140] T. Jiao, Y. Ma, X. Guo, Y. Ye, C. Xie, Bile acid and receptors: biology and drug discovery for nonalcoholic fatty liver disease, Acta Pharm. Sin. 43 (5) (2022) 1103–1119, https://doi.org/10.1038/s41401-022-00880-z.

[141] B. Lakić, R. Akrbić, S. Uletilović, N. Mandić-Kovačević, M. Grabež, M.P. Aarić, M. P. Stojiljković, I. Soldatović, Z. Janjetović, A. Stokanović, N. Stojaković, M. Mikov, Beneficial effects of ursodeoxycholic acid on metabolic parameters and oxidative stress in patients with type 2 diabetes mellitus: a randomized double-blind, placebo-controlled clinical study, J. Diabetes Res 2024 (2024) 4187796, https://doi.org/10.1155/2024/4187796.

[142] S.I. Stan, V. Biciuşcă, D. Clenciu, A. Mitrea, M. Boldeanu, P. Durand, S. Dănoiu, The therapeutic mechanisms and beneficial effects of ursodeoxycholic acid in the treatment of nonalcoholic fatty liver disease: a systematic review, Med Pharm. Rep. 97 (1) (2024) 12–25, https://doi.org/10.15386/mpr-2629.

[143] S.K. Hourigan, M. Ahn, K.M. Gibson, M. Pérez-Losada, G. Felix, M. Weidner, I. Leibowitz, J.E. Niederhuber, C.L. Sears, K.A. Crandall, M. Oliva-Hemker, Fecal transplant in children with clostridioides difficile gives sustained reduction in antimicrobial resistance and potential pathogen burden, Open Forum Infect. Dis. 6 (10) (2019), https://doi.org/10.1093/ofid/ofz379.

[144] E.D. Hernandez, L. Zheng, Y. Kim, B. Fang, B. Liu, R.A. Valdez, W.F. Dietrich, P. V. Rucker, D. Chianelli, J. Schmeits, D. Bao, J. Zoll, C. Dubois, G.C. Federe, L. Chen, S.B. Joseph, L.B. Klickstein, J. Walker, V. Molteni, P. Mcnamara, S. Meeusen, D.C. Tully, M.K. Badman, J. Xu, B. Laffitte, Tropifexor-mediated abrogation of steatohepatitis and fibrosis is associated with the antioxidative gene expression profile in rodents, Hepatol. Commun. 3 (8) (2019) 1085–1097, https://doi.org/10.1002/hep4.1368.

[145] A. Medford, J. Childs, A. Little, S. Chakraborty, L. Baiocchi, G. Alpini, S. Glaser, Emerging therapeutic strategies in the fight against primary biliary cholangitis, J. Clin. Transl. Hepatol. 11 (4) (2023) 949–957, https://doi.org/10.14218/JCTH.2022.00398.